

Integrated optimal operation of power and water systems under uncertainty: An adjustable robust optimization approach

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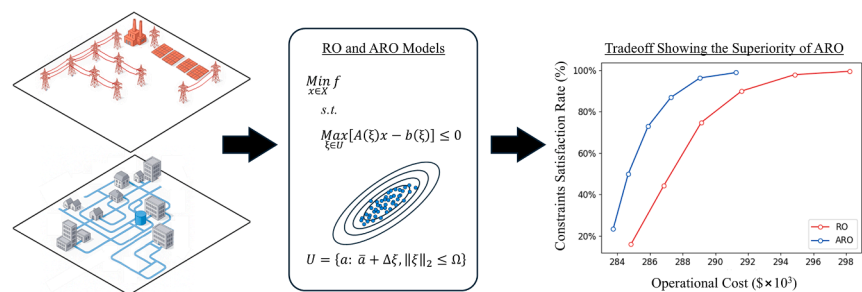
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HIGHLIGHTS

- We propose a non-probabilistic RO and ARO framework for integrated power-water operation under uncertainty.
- Introduced a general method to eliminate uncertain equality constraints for tractable RO and ARO.
- We demonstrate how decision recourse in ARO can provide superiority over static RO in terms of cost-robustness tradeoff.
- We demonstrate how ARO can accommodate real-time adjustability for different scenarios.
- The price of robustness reveals that robustness can be substantially improved over deterministic optimization with only a limited increase in operational costs.

GRAPHICAL ABSTRACT



ARTICLE INFO

Keywords:

Adjustable robust optimization
Real-time
Data-driven
Integrated systems
Uncertain equality constraints

ABSTRACT

Water and power systems are deeply interdependent, yet they are typically analyzed and optimized separately. Coupling these systems leverages their mutual dependency to enhance overall efficiency and reliability, although it introduces significant modeling complexity, especially when considering the inherent uncertainty in these systems, such as water demands, power loads, and renewable generation. This study proposes Robust Optimization (RO) and adjustable robust optimization (ARO) for the integrated day-ahead scheduling of pump operations and generator dispatch under uncertainty. A novel formulation is introduced to systematically eliminate uncertain equality constraints, which hinders the direct implementation of RO and ARO models. This is done by decomposing the decision vector into dependent and independent variables and substituting the dependent variables to eliminate the equality constraints. A key advantage of the ARO framework is its ability to incorporate real-time information as uncertainty unfolds, enabling adaptive decision-making. By coupling the two systems, the model allows each system to adjust its operation not only based on internal measurements but also in response to the evolving state of the other system. Two case studies demonstrate the method's performance through a Pareto tradeoff between cost optimality and robustness. The results show that robustness can be

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<https://doi.org/10.1016/j.watres.2026.125325>

Received 2 July 2025; Received in revised form 24 December 2025; Accepted 1 January 2026

Available online 2 January 2026

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dramatically improved, compared to deterministic optimization, increasing the constraint satisfaction rates from less than 25% to above 99% with an increase of only 2-5% in operational costs. The results also highlight the superiority of ARO over RO in achieving lower operational costs for comparable levels of robustness. These findings showcase the potential of the proposed method to generate robust, adaptive policies that support the advancement of sustainable and reliable operation of interdependent infrastructure systems.

Nomenclature

Notation

$\mathcal{B}$	Set of all buses
$\mathcal{G}$	Set of all generators
$\mathcal{E}$	Set of all batteries
pc	Pump combination (a subset of pumps in a pumping station operated simultaneously)
$\mathcal{D}$	Set of all desalination plants
K	Number of piecewise linear segments
$\Theta_{pc}(i)$	Set of all pump combinations connected to bus i
$\Theta_D(i)$	Set of all desalination plants connected to bus i
$\mathcal{S}$	Set of all storage tanks
n	Number of buses in the system
T	Number of time steps
t	Index of time steps
Δt	Time step size
I	Set of independent decision variables
D	Set of dependent decision variables
U	Uncertainty set
Ω	Ellipsoid radius
Δ	A mapping matrix that defines the uncertainty set; computed from a covariance matrix Σ
Λ	A matrix represents the non-anticipativity pattern

Optimization Decision Variables

$\theta_{i,t}$	Voltage angle at bus i and time t
$w_{i,t}$	Auxiliary variables for the piecewise linear approximation of the cost function for generator i at time t
$x_{i,t}^g$	Active power generated at bus i and time t
$x_{i,t}^{bat}$	Battery charging of battery at bus i and time t (negative for discharging)
$x_{pc,t}^p$	Portion of time step where the pump combination pc is activated at time t
$x_{d,t}^D$	Volume of desalinated water at desalination plant d and

time t

$x_{s,t}^{tank}$	Flow into tank s at time t (negative for outflow)
$SOC_{i,t}$	State of charge of battery at bus i and time t
$V_{s,t}$	Volume of tank s at time t
π_0, π_1	LDR coefficients (i.e., decision variables) for the ARO formulation representing
ξ	Uncertain variable representing fluctuations from nominal values

Parameters

a_i, b_i, c_i	Coefficients of the quadratic cost function of generator i
$\alpha_{i,k}, \beta_{i,k}$	Coefficients that define the k^{th} linear segment of the cost function of generator i
$L_{i,t}$	Active power load at bus i and time t
$PV_{i,t}$	Renewable power injection at bus i and time t
$\underline{\theta}, \bar{\theta}$	Min and max bounds of the voltage angle
$\underline{x}_i^g, \bar{x}_i^g$	Min and max bounds of the power generation at generator i
$\underline{x}_i^{bat}, \bar{x}_i^{bat}$	Min and max bounds of the battery charging/discharging rate at bus i
$\underline{x}_k^d, \bar{x}_k^d$	Min and max bounds of the desalination production at desalination plant k
$SOC_i, \bar{SOC}_i$	Min and max bounds of the state of charge of the battery at bus i
$\underline{V}_s, \bar{V}_s$	Min and max bounds for water volume at tank s
$SOC_{i,init}, SOC_{i,final}$	Initial and final state of charge of the battery at bus i
$V_{s,init}, V_{s,final}$	Initial and final water volume at tank s
R_i	Max up/down ramp rate of generator i
B_{ij}	Element of susceptance between bus i and bus j
q_{pc}	Nominal flow of pumps combination pc
P_{pc}	Nominal power of pumps combination pc
P_d	Power (per 1 m ³ of water) required for the desalination plant d
$d_{s,t}$	Water demand from tank s at time t

1. Introduction

In recent years, there has been growing interest in understanding and leveraging the interdependencies between infrastructure systems, driven by the belief that integrated decision-making can enhance the overall efficiency and resilience of these systems (Coskun Dilcan and Aydinalp Koksall, 2025; Yin et al., 2023). A prominent example is the interconnection between Power Distribution Systems (PDS) and Water Distribution Systems (WDS) (Molinis-Senante et al., 2022). These two systems exhibit a bidirectional dependency: energy-intensive facilities in the WDS such as pumping stations, desalination, and treatment plants, constitute significant power loads. At the same time, WDS provides considerable operational flexibility through storage facilities, allowing load-shifting capabilities, which are essential for managing the dynamic demands of PDS (Sowby, 2023). This study focuses on an interesting case of integrated systems involving large, high-voltage systems that supply power to desalination plants and large pumping stations. In such systems, the power loads consumed by the water system are relatively

large and constitute a significant portion of the total generated loads (Wu et al., 2024), thus highlighting the importance of coupled decision-making and coordination between the water and power systems (Portilla-Paveri et al., 2024; Toba et al., 2024).

Current global trends pose significant challenges to both PDS and WDS management: demand for both water and energy is generally increasing, while the variability in demand patterns and the availability of the resources to meet it, becoming increasingly uncertain. For example, in certain regions, rapid urbanization, population growth, and economic development are driving increased demand for both electricity and water. Additionally, the increasing scarcity of accessible freshwater in many regions forces a shift toward more energy-intensive sources, such as seawater desalination and pumps that extract water from increasingly deeper and more distant aquifers, thereby elevating energy requirements by WDS. A major concern of decision-makers in PDS and WDS is the multifaceted uncertainty of these systems. The

demands for power and water are the main sources of uncertainty. The availability of water and power sources is also uncertain, where climate change affects the water sources' availability patterns (Szinaï et al., 2024), while power networks are affected by the high penetration rates of variable renewable energy sources, such as solar panels and wind turbines, introducing intermittency into the power supply (Jiang et al., 2025). Addressing these challenges requires the development of new methods that generate efficient and robust operational policies.

The optimization of PDS and WDS operations as individual entities is a mature field of research with numerous studies published over the last decades, spanning various modeling and solution approaches. For instance, economic dispatch in PDS has been studied under deterministic assumptions (Gao et al., 2024; Wang and Xiong, 2025; Wen et al., 2024) as well as under uncertainty (Basak et al., 2023; Chacko et al., 2024; Nourianfar and Abdi, 2023). Similar research has been conducted in the WDS analysis community, covering both deterministic (Brás et al., 2025; Creaco et al., 2019; Salomons and Housh, 2020a, 2020b; Thomas and Sela, 2024), and uncertain formulations, such as stochastic optimization (Grosso et al., 2017; Rafael Schwartz et al., 2016) and scenario-based optimization (Salomons et al., 2025).

When optimizing WDS independently of the connected PDS, it is assumed that the electricity tariffs are static and determined offline based on an analysis made by the power utility to incentivize heavy consumers to shift loads to off-peak times. However, such measures still fall short of capturing the full complexity and day-to-day dynamics of system operations. Even though tariffs may distinguish between off-peak and on-peak pricing, their static nature prevents them from adapting to daily fluctuations in weather, consumption, and other dynamic factors, ultimately leading to suboptimal operational decisions. With the growing trend of real-time energy pricing, the need for dynamic adjustment becomes even more critical, as near-future conditions remain uncertain and rapidly evolving. By integrating the optimization of PDS and WDS, dynamic adjustability is achieved. In such frameworks, WDS loads are no longer triggered by static tariffs but instead adapt in real-time to the daily fluctuations, ultimately leading to more efficient operations.

Following this insight, there has been a growing interest in the realm of integrated PDS-WDS optimization. For instance, several studies proposed optimal power-water flow models that demonstrated improved overall efficiency compared to independent optimization (Naval and Yusta, 2021; Sharma and Li, 2024; Shmaya et al., 2024). The impact of the PDS modeling approach was investigated by Alhazmi and Dehghanian (2021), who compared DC power flow with linearized AC power flow. In both cases, WDS modeling was based on piecewise linearization. The results show that the overall costs of joint optimization are superior to individual systems optimization. Desalination plants have also received attention in the integrated optimization literature due to their significant electricity demands. In these scenarios, the WDS accounts for a larger share of the total system load, thereby amplifying the benefits of coordinated decision-making. A mixed integer linear programming approach was suggested by Alnahhal et al. (2023), utilizing flexibility in desalination energy recovery to reduce the overall costs of the integrated systems. A similar study by Oikonomou and Parvania (2020) highlighted the advantages of joint optimization, demonstrating its superiority compared to WDS's independent optimization under different electricity tariff structures.

1.1. Integrated optimal operation under uncertainty

The above studies lay the foundation for integrated optimization, with a predominant focus on deterministic scenarios. However, there is still a gap in applying these models to large-scale, real-world conditions that involve various intertwined uncertain factors, such as water demands, power loads, and renewable energy availability, which all correlate with climatic uncertainty. Only a limited number of studies addressed these uncertainties, intending to improve not only the

operational costs but also the robustness of optimal policies. For example, Stuhlmacher and Mathieu (2020a) adopted a chance constraint (CC) method to optimize integrated PDS-WDS under power load uncertainty. A follow-up study extended this approach to include uncertainty in both power loads and water demands (Stuhlmacher and Mathieu, 2020b). A similar approach was suggested by Alhazmi et al. (2023), in which joint probabilistic constraints were used to consider uncertainty in renewable availability and water demands. Zhao et al. (2024) presented another CC approach considering uncertainties in power loads, water demands, and renewable energy availability. Their proposed solution strategy is based on a coupled algorithm including iterative deterministic solutions with a deep neural network to simulate uncertain scenarios. Another recently presented stochastic approach is a stochastic predict-then-optimize framework (Wu et al., 2025), which integrates probabilistic forecasting with stochastic optimization through a differentiable regret-based loss.

Despite their contributions, these studies rely primarily on classical stochastic optimization techniques. While effective in modeling uncertainty when probability distributions are known, such approaches pose two key limitations. First, they often lead to nonlinear and computationally intractable formulations, especially in large-scale settings (Ben-Tal et al., 2009; Birge and Louveaux, 2011). Second, they depend on the availability of accurate probability distribution functions (PDF), which are often unavailable or unreliable in real-world contexts characterized by deep uncertainty, such as those influenced by climate change (Maier et al., 2016).

Alternatively, the non-probabilistic Robust Optimization (RO), introduced by Ben-Tal and Nemirovski (1999) offers a tractable formulation that ensures the feasibility of the optimal policy against a range of scenarios defined within uncertainty sets. A prominent advantage of RO is its ability to define uncertainty sets based on partial probabilistic information (such as the first two statistical moments), sparing the need for an exact PDF. RO was successfully used in the context of water resources planning and management (Housh, 2017; Housh et al., 2011; Pankaj et al., 2022; Perelman et al., 2023, 2013; Refael Schwartz et al., 2016), and in power systems (Molzahn and Roald, 2018; Xie et al., 2024).

The underlying probability distribution itself can be treated as uncertain, an approach commonly referred to as distributionally robust optimization (DRO). DRO has been explored in integrated infrastructure systems (Zhou et al., 2025), but its advantages are less relevant to the present setting. The uncertainties considered here of forecast errors in power loads, water demands, and PV generation are well approximated by centered and approximately Gaussian distributions. In such cases, the ambiguity in the distribution is limited, and DRO would introduce additional computational complexity without offering commensurate benefits.

For multistage decision making, the dynamic version of RO, known as the Adjustable Robust Optimization (ARO) (Ben-Tal et al., 2004), enables adaptive decision-making by modeling the sequential nature of the decision-making process in multi-period problems. Specifically, at each period t , the uncertainty realized in earlier periods is already known; thus, the decision at t can adjust accordingly. This is achieved by defining the decision variables as affine functions of past observed values, known in the literature as the Affine ARO (AARO), the reader is referred to Ben-Tal et al. (2004) for more details.

The advantages of ARO were demonstrated in several applications in both power (Jabr, 2013; Tan et al., 2025; Yan et al., 2022) and water systems (Housh et al., 2012; Perelman and Ostfeld, 2023). Based on these studies, we hypothesize that ARO can also have advantages in coupled power water systems compared to other methods. In this context, the adjustable formulation facilitates dynamic decision-making by allowing both systems to respond adaptively to evolving conditions. Through the affine decision rules, water-system decisions (e.g., pumping and desalination) can adjust to the realized states of the power system, and vice versa, achieving cross-system synergistic optimization that

enhances both robustness and cost efficiency.

Stuhlmacher and Mathieu (2022) suggested an ARO approach for integrated PDS-WDS focused on voltage support optimization. In their study, the pumps' power is modeled as an affine policy depending on the forecasting error of the same time step. While this approach may be acceptable for high-resolution processes like voltage control, for economic dispatch, the forecasting error is not available at the time of the decision. Moreover, the affine policy suggested in Stuhlmacher and Mathieu (2022) does not utilize the uncovered uncertain values from previous periods that are likely to be available at the time of the decision. Based on the assumption that time steps are correlated, including the measured values from the near past can significantly enhance the affine policy efficiency.

While RO and ARO hold a valuable potential to cope with the challenge of optimizing integrated PDS-WDS under uncertainty, a significant challenge arises when incorporating equality constraints in these methods. In the following section, we present this challenge and review approaches for handling equality constraints in RO and ARO models and motivate the strategy adopted in this work.

1.2. Challenge of equality constraints

According to Ben-Tal et al. (2009), when formulating RO and ARO, it is advisable to avoid equality constraints when uncertainty is present. The source of this challenge is that robust solutions must satisfy all scenarios within a predefined uncertainty set. In inequality constraints, this can be achieved by finding the worst-case scenario, which, if satisfied by the optimal solution, will be feasible for all other scenarios. However, in the case of an equality constraint, it can only satisfy a single realization of the uncertain parameters. Consider, for example, the following equality constraint, $x_1 + x_2 = \xi$, where ξ is an uncertain parameter; this equation cannot be satisfied for two different realizations of ξ with the same solution of x_1 and x_2 .

Nonetheless, equality constraints are essential for modeling physical conservation laws, such as power balance in PDS or mass balance in WDS. Consequently, several strategies have been proposed in the literature to address this issue: One straightforward approach is to relax equality constraints by converting them into appropriate inequalities. For example, based on the assumption that excess energy in PDS can be curtailed at no cost, Keyvandarian and Saif (2024) replaced power balance equations with inequality constraints at demand nodes. Although such relaxation does not preserve an equivalent feasible region. Nevertheless, when solving a minimization problem with strictly positive objective coefficients, any excess beyond the required energy naturally increases the objective value. Thus, the optimal solution inherently tends toward binding constraints that retain the equality constraints. Alternatively, some studies model the power balance as deterministic constraints using nominal values, embedding uncertainty only in the cost function and inequality constraints. This approach has been used for ARO (Jabr, 2013) and distributionally robust optimization (DRO) (Mieth and Dvorkin, 2018). A more rigorous approach is adopted by Bertsimas et al. (2013), who explicitly replace each equality with two robust inequalities. In their formulation, both sides of these inequalities employ adaptive affine policies that are directly dependent on uncertain parameters, thereby guaranteeing robust satisfaction of equality under all uncertainty realizations. However, these approaches entail certain drawbacks. They either partially simplify the uncertainty representation by neglecting it in specific constraints or impose strict robust inequalities, significantly narrowing the feasible region and potentially increasing the conservativeness of solutions.

Another strategy is to use the equalities to eliminate variables, as suggested by Ben-Tal et al. (2009) and discussed by Gorissen et al. (2015). The idea is to manifest some variables as functions of others based on the equality constraint equations, then substitute the eliminated variables everywhere else in the problem. Naturally, this technique results in a decomposition into dependent and independent

variables. In the case of state equations, this decomposition is intuitive. For example, in WDS, the mass balance at tanks can be formulated using the state equation as in Eq. (1).

$$V_t = V_{t-1} + Q_{in,t} - Q_{out,t}, \quad (1)$$

where V_t is the tank volume at time t , and $Q_{in,t}$, $Q_{out,t}$ are the inflow and outflow from the tank, respectively. Often, the state variable, V_t , is bounded as given in Eq. (2).

$$\underline{V} \leq V_t \leq \bar{V} \quad \forall t \in \{1 \dots T\}, \quad (2)$$

To eliminate the state variables, Ben-Tal et al. (2009) suggested expressing the volume at time t as the cumulative sum of inflows and outflows up to time t , and formulating the inequality constraints based on this sum, Eq. (3).

$$\underline{V} \leq \sum_{r=1}^t (Q_{in,r} - Q_{out,r}) \leq \bar{V} \quad \forall t \in \{1 \dots T\}, \quad (3)$$

This approach was implemented for RO (Perelman et al., 2023) and ARO (Goryashko and Nemirovski, 2014; Perelman and Ostfeld, 2023) for WDS optimization. Nevertheless, unlike state equations, from which state variables could be trivially eliminated, in equality constraints that involve different decision variables, the elimination of decision variables is less trivial since there is no clear distinction between dependent and independent variables. For example, consider the mass balance in the WDS junction as given in Eq. (4). It is unclear which of the flows should be the dependent flows, and which are the independent flows.

$$\sum_{i \in J_{in,j}} Q_{i,j} - \sum_{i \in J_{out,j}} Q_{j,i} = d_j \quad \forall j \quad (4)$$

The aforementioned elimination methods neither simplify the uncertainty modeling nor significantly constrain the feasible region. Nonetheless, it has primarily been demonstrated using cases involving intuitive state variable elimination. Thus, there remains a need for a systematic and generalizable methodology for conducting this elimination. To address this critical gap, this paper proposes a generalized and systematic method for decomposing equality-constrained in RO and ARO problems.

1.3. Contributions

This study develops a tractable optimization framework for the day-ahead operation of large-scale integrated power and water systems under uncertainty. The focus is on national or regional-scale infrastructures that involve high-energy consumption pumping (in the MW range), major lines, and desalination plants that constitute a significant share of the total power grid loads and thus necessitate coordinated decision-making. The proposed formulation adopts linearized models (DC power flow and linear pump scheduling) that are consistent with the regional scale of the system and the planning horizon, where decisions concern aggregate dispatch, pumping, and storage rather than detailed hydraulics and power flow in the system. Among the various strategies for modeling uncertainty, we employ ellipsoidal sets within RO and ARO frameworks. This choice aligns with the statistical characteristics of the forecast errors considered in this study, which are centered, bounded, and approximately symmetric. The ellipsoidal RO, ARO formulation provides an accurate and computationally efficient representation of these uncertainties, while the ARO structure enables adaptive recourse actions that are essential for coordinated water–power operation across the day-ahead horizon. The main contributions are as follows:

- Develop a generalized and systematic method for eliminating uncertain equality constraints in RO and ARO formulations, enabling the robust optimization of coupled power-water systems without resorting to ad-hoc relaxations.

- Formulate RO and ARO models for the integrated PDS-WDS optimal operation that remain computationally tractable for large-scale day-ahead applications while capturing the key interdependencies between the two systems.
- Analyze how different selections of dependent and independent variable sets affect the resulting optimization problems.
- Demonstrate the performance of RO and ARO through two case studies, quantifying the tradeoff between cost optimality and robustness, and highlighting the superiority of ARO in achieving lower cost with fewer constraint violations.

2. Methodology

We solve an integrated day-ahead optimization of coupled PDS-WDS under uncertainty of power loads, water demands, and PV availability. The objective is to minimize the power generation cost while satisfying water demand and power loads. The PDS examined in this study consists of conventional generators, renewable sources, and storage facilities (e. g., batteries). The PDS also supplies power to a connected WDS, which consists of desalination plants, pumping stations, and storage tanks. With an emphasis on traceability, the optimization constraints are formulated based on several relaxations. Power flow simulation is traditionally performed using the Newton-Raphson method, known for its accuracy but also for its high computational complexity (Irving and Sterling, 1987; Tinney and Hart, 1967), which poses significant challenges for our purposes. To address this challenge, we use DC power flow (DCPF) approximation, which is widely adopted as an efficient approximation method for power flow analysis in power systems (Taheri and Molzahn, 2024). The WDS model is based on an implicit description of the system hydraulics. By using pre-computed system states, the connectivity and energy consumption of pumps are translated into linear equations, relaxing the optimization effort (Jowitt and Germanopoulos, 1992; Ormsbee and Lansey, 1994). The result of these approximations is a linear program that describes the deterministic case of the integrated PDS-WDS optimal operation as presented in Eqs. (5)–(21), where the introduction of uncertainties and robustification of the problem can transform the problem into a second-order conic program, as explained below. Recall that the focus of this study is on national-scale, day-ahead operation, hence the linear modeling assumptions are both reasonable and common. The use of a DC power-flow (DCPF) approximation for the power grid and linear pump-scheduling relations for the water system provides a sufficiently accurate yet computationally tractable representation of the dominant interactions between the two infrastructures.

The objective function is to minimize the overall power generation cost, commonly described by quadratic functions of the generated power as presented in Eq. (5).

$$\min \sum_{t=1}^T \sum_{i \in G} \left[a_i \cdot (x_{i,t}^g)^2 + b_i \cdot x_{i,t}^g + c_i \right], \quad (5)$$

where G is the set of all generators, $x_{i,t}^g$ is the power generated at bus i at time t . To maintain the linearity of the deterministic problem, the quadratic cost function is converted to a convex piecewise linear function with K segments. Owing to its convexity, based on formulating the epigraph of Eq. (5), the piecewise linear function can be modeled using a set of linear constraints by introducing auxiliary decision variables $w_{i,t}$ as described in Eq. (6) (Coffrin et al., 2021).

$$a_{i,k} \cdot x_{i,t}^g + \beta_{i,k} \leq w_{i,t} \quad \forall i \in G, \quad \forall k \in \{1 \dots K\}, \quad \forall t \in \{1 \dots T\}, \quad (6)$$

where $a_{i,k}$ and $\beta_{i,k}$ are the coefficients that define the k^{th} linear segment of generator i , and $w_{i,t}$ is an auxiliary variable for the i^{th} generator at time t . Accordingly, the piecewise linear objective function is given by Eq. (7).

$$\min \sum_{t=1}^T \sum_{i \in G} w_{i,t}, \quad (7)$$

The DCPF model is formulated with power balance at every bus as presented in Eq. (8). The power loads and PV injections are functions of the uncertain variable ξ , which resides in the uncertainty set U , and represents the uncertain fluctuations in the uncertain factors of the problem.

$$x_{i,t}^g + \sum_{j=1}^n B_{ij} \theta_{j,t} - x_{i,t}^{\text{bat}} - \sum_{pc \in \Theta_{pc}(i)} P_{pc} x_{pc,t}^p - \sum_{d \in \Theta_D(i)} P_d x_{d,t}^D = L_{i,t}(\xi) - PV_{i,t}(\xi)$$

$$\forall i \in \mathcal{B}, \quad \forall t \in \{1 \dots T\}, \quad (8)$$

where $x_{i,t}^g$ is the power generated at bus i at time t . B_{ij} is the susceptance between bus i and j . $\theta_{j,t}$ is the voltage angle of bus j at time t . Θ with index denotes the set of devices connected to bus i (pumps or desalination plants). P_{pc} is the nominal power consumed by the operation of pump combination pc , and $x_{pc,t}^p$ is the portion of time step t where pump combination pc is activated. P_d is the specific power (power per volume of water) consumed by desalination plant d , and $x_{d,t}^D$ is the volume of water desalinated by desalination plant d at time t . $L_{i,t}(\xi)$ and $PV_{i,t}(\xi)$ are uncertain power load and renewable power injection in bus i at time t . $\mathcal{B}$ is the set of all buses, and T is the number of time steps.

The voltage angle of a reference bus, denoted here as bus 1, equals zero at all time steps, Eq. (9)

$$\theta_{1,t} = 0 \quad \forall t \in \{1 \dots T\}, \quad (9)$$

The state of charge (SOC) of batteries is between min and max capacities, Eq. (10). The representation of the SOC at each time step is based on a cumulative sum of charging and discharging from the first time step. This is done to support state variable elimination, see the discussion above regarding the challenges of uncertain equality constraint modeling.

$$\underline{SOC}_i \leq SOC_{i,\text{init}} + \sum_{r=1}^t x_{i,r}^{\text{bat}} \leq \overline{SOC}_i \quad \forall i \in E, \quad \forall t \in \{1 \dots T\}, \quad (10)$$

where $SOC_{i,t}$ is the SOC of a battery connected to bus i at time t . $x_{i,r}^{\text{bat}}$ is the battery charging (negative for discharging) at time r . $\underline{SOC}_i$ and $\overline{SOC}_i$ are the lower and upper bounds of the battery SOC. E is the set of all batteries.

To maintain a reasonable starting point for the next day, the SOC at the end of the current day becomes the lower bound of the initial SOC for the following day, effectively carrying it over to cope with the daily operational cycle, Eq. (11).

$$SOC_{i,\text{final}} \leq SOC_{i,\text{init}} + \sum_{t=1}^T x_{i,t}^{\text{bat}} \quad \forall i \in E, \quad (11)$$

The variation in power generation over consecutive time steps is limited by a ramp rate, Eq. (12).

$$|x_{i,t}^g - x_{i,t-1}^g| \leq R_i \quad \forall i \in G, \quad \forall t \in \{1 \dots T\}, \quad (12)$$

where $x_{i,t}^g$ is the power generated at bus i at time t . R_i is the ramping bound for the generator connected to bus i . G is the set of all generators.

The bus voltage angles, power generation, and battery charging must be within their lower and upper bounds, see Eqs. (13)–(15) respectively.

$$\underline{\theta} \leq \theta_t \leq \bar{\theta} \quad \forall t \in \{1 \dots T\}, \quad (13)$$

$$\underline{x}_i^g \leq x_{i,t}^g \leq \bar{x}_i^g \quad \forall i \in G, \quad \forall t \in \{1 \dots T\}, \quad (14)$$

$$\underline{x}_i^{\text{bat}} \leq x_{i,t}^{\text{bat}} \leq \bar{x}_i^{\text{bat}} \quad \forall i \in E, \quad \forall t \in \{1 \dots T\}, \quad (15)$$

where $\underline{\theta}$, $\bar{\theta}$ are the lower and upper bounds of the voltage angles. Similarly, $\underline{x}_i^g$, $\bar{x}_i^g$ are the lower and upper dispatch bounds of the generator connected to bus i , and $\underline{x}_i^{bat}$, $\bar{x}_i^{bat}$ are the lower and upper charging (and discharging) bounds the battery connected to bus i .

The WDS modeling is also based on a set of constraints. First, the water balance at each pressure zone is maintained, where a pressure zone is governed by a tank and characterized by inflows and outflows, Eq. (16). The water demand is a function of the uncertainty variables ξ , similar to the power loads and PV injections in the bus power balance.

$$\sum_{d \in S_{in,s}} x_{d,t}^D + \sum_{pc \in S_{in,s}} q_{pc} x_{pc,t}^p - \sum_{pc \in S_{out,s}} q_{pc} x_{pc,t}^p - x_{s,t}^{tank} = d_{s,t}(\xi) \quad \forall s \in S, \quad \forall t \in \{1 \dots T\}, \quad (16)$$

where $x_{d,t}^D$ is the volume of water desalinated by desalination plant d at time t . q_{pc} is the nominal flow of pump combination pc , and $x_{pc,t}^p$ is the portion of time step t where pump combination pc is activated. $x_{s,t}^{tank}$ is the inflow to tank s at time t . $d_{s,t}(\xi)$ is the uncertain water demand from tank s at time t . S is the set of all water tanks.

Similarly to the batteries, the tank volumes are bounded to be between min and max volumes, and the final volumes are larger or equal to the initial volumes, see Eqs. (17) and (18) respectively.

$$\underline{V}_s \leq V_{s,init} + \sum_{r=1}^t x_{s,r}^{tank} \leq \bar{V}_s \quad \forall s \in S, \quad \forall t \in \{1 \dots T\}, \quad (17)$$

$$V_{s,final} \leq V_{s,init} + \sum_{t=1}^T x_{s,t}^{tank} \quad \forall s \in S, \quad (18)$$

where $V_{s,t}$ is the water volume in tank s at time t . $x_{s,t}^{tank}$ is the inflow to tank s at time t . $\underline{V}_s$, $\bar{V}_s$ are the lower and upper bounds of tank s volume.

The variables $x_{pc,t}^p$ indicate the fraction of time when pump combination pc is activated out of a time step with duration Δt . Therefore $x_{pc,t}^p$ must be larger than 0, and the sum of all activated pump combinations of a pumping station at any time step must be between 0 and Δt , see Eqs. (19) and (20).

$$\sum_{pc \in PC_{ps}} x_{pc,t}^p \leq \Delta t \quad \forall ps \in PS, \quad \forall t \in \{1 \dots T\}, \quad (19)$$

$$0 \leq x_{pc,t}^p \quad \forall pc \in PC_{ps}, \quad \forall ps \in PS, \quad \forall t \in \{1 \dots T\}, \quad (20)$$

Lastly, the desalination flows must be between lower and upper bounds, Eq. (21).

$$\underline{x}_d^D \leq x_{d,t}^D \leq \bar{x}_d^D \quad \forall d \in D, \quad \forall t \in \{1 \dots T\}, \quad (21)$$

where $x_{d,t}^D$ is the volume of water desalinated by the desalination plant d at time t , and $\underline{x}_d^D$, $\bar{x}_d^D$ are the lower and upper bounds of the desalination volume in the desalination plant d .

As noted above, the formulation involves several uncertain parameters notated as a function of an uncertain variable $\xi \in U$. It is assumed that an uncertain parameter, a , can be described as: $a = \bar{a} + \Delta \xi$, where $\bar{a}$ indicates the known nominal value, Δ is a mapping matrix that governs the uncertainty properties based on a covariance matrix (Ben-Tal et al., 2009), and ξ represents the uncertain fluctuations from the nominal value.

For convenience of notation, the above formulation can be succinctly written in matrix form as shown in Eq. (22). This matrix form introduces no new assumptions but serves only as a compact representation of the physical constraints defined in Eqs. (5)–(21). Two comments worth noting about the matrix formulation: 1) To have all uncertain expressions in the constraints set, the objective function is modeled as a constraint using an auxiliary decision variable f , and an inequality constraint is added such that $\sum_{t=1}^T \sum_{i \in G} w_{i,t} \leq f$; 2) It is noted that the

model in Eqs. (5)–(21) has a special property in which all the uncertain parameters are present only in the equality constraints right hand side. This allows us to keep the vector c deterministic, which will be helpful in later stages.

$$\min_x f$$

s.t.:

$$Ax = b(\xi)$$

$$Bx \leq c \quad (22)$$

where x is a vector of decision variables that collects all the decision variables in Eqs. (5)–(21). For ease of notation and without loss of generality, f marks the last element of the vector x . A , B , and c are deterministic parameters that determine the networks' connectivity, operational constraints, lower, and upper bounds; $b(\xi)$ is the RHS vector of the equality constraints, which includes the uncertain loads, PV injections, and water demands. Note that, without loss of generality, some elements of the vector b can be independent of ξ .

2.1. Equality constraints elimination

The linear system in the equality constraint is an underdetermined linear system, as we have more decision variables than equality constraints in our problem. This is as a result of the mass balance equation in the water system, as well as power balance equations in the optimal DC power flow formulation, in which the number of equality constraints corresponds only to the number of buses, whereas the decision variables include bus voltage angles (one per bus), generators dispatch, and batteries charging rates. Consequently, if A is $m \times n$ matrix, then $m < n$ and A is underdetermined. When A has a full row rank, then the underdetermined system has an infinite number of solutions, which can be characterized using a decomposition of a linear system by selecting m independent columns of A and their corresponding variables from the overall decision vector x . Noteworthy that the elements of x which correspond to independent columns of A can be considered dependent decision variables since they can be uniquely determined for given values of elements in x which correspond to dependent columns of A . Thus, hereinafter, we use the descriptive *dependent/independent* with respect to the decision vector, and not to the column space of A .

Using the decomposition to dependent and independent decision variables, the general solution of the underdetermined system is shown in Eq. (23), where I and $\mathcal{D}$ represent the sets of indices corresponding to independent and dependent decision variables, respectively.

$$A = [A_{\mathcal{D}}, A_I], \quad x = [x_{\mathcal{D}}, x_I]$$

$$A_{\mathcal{D}} x_{\mathcal{D}} + A_I x_I = b(\xi)$$

$$x_{\mathcal{D}} = A_{\mathcal{D}}^{-1} [b(\xi) - A_I x_I] \quad (23)$$

As pointed out by Housh (2011), the decomposition itself can be done in several ways. For example, in graph theory, finding the dependent columns of a matrix is equivalent to finding the non-spanning edges of a graph, where in a connected graph, there may be many spanning trees, and any of them can be used. Other decomposition methods can be finding the Null space of A or the reduced row Echelon form (RREF), where pivot columns are independent. Once A is decomposed, we seek to reconstruct a vector $x \in \mathbb{R}^n$ in a structured form to facilitate further analysis and maintain consistency with the original formulation. To achieve this, we introduce two partial permutation matrices $p_I \in \mathbb{R}^{n \times |I|}$, $p_{\mathcal{D}} \in \mathbb{R}^{n \times |\mathcal{D}|}$ defined as in Eq. (24),

$$p_I[I, :] = \text{Identity}(|I|)$$

$$p_{\mathcal{D}}[\mathcal{D}, :] = \text{Identity}(|\mathcal{D}|) \quad (24)$$

where $Identity(m)$ denotes the identity matrix of m . The notation $p_I[I, :]$ implies the collection of all rows of p_I in the index set I . Using these matrices and the general solution for the dependent variables, the entire decision variable vector can be compactly written as Eq. (25).

$$x = p_{\mathcal{D}} x_{\mathcal{D}} + p_I x_I$$

$$x = p_{\mathcal{D}} A_{\mathcal{D}}^{-1} [b(\xi) - A_I x_I] + p_I x_I \quad (25)$$

The next step is substituting Eq. (25) in the inequality constraints from Eq. (22).

$$\begin{aligned} \min_{x_I} f \\ \text{s.t.:} \\ B[p_{\mathcal{D}} A_{\mathcal{D}}^{-1} (b(\xi) - A_I x_I) + p_I x_I] \leq c \end{aligned} \quad (26)$$

For ease of notation, we define $k_1 = p_{\mathcal{D}} A_{\mathcal{D}}^{-1}$ and $k_2 = p_I - p_{\mathcal{D}} A_{\mathcal{D}}^{-1} A_I$, thus we can write the optimization problem as given in Eq. (27), in which we are left with only inequality constraints involving uncertain parameters only in the RHS of the constraint.

$$\begin{aligned} \min_{x_I} f \\ \text{s.t.:} \\ Bk_2 x_I \leq c - Bk_1 b(\xi) \end{aligned} \quad (27)$$

2.2. Robust counterpart

The robustification of the problem in Eq. (27) involves maximizing the uncertain parameters over a predefined uncertainty set (Ben-Tal et al., 2009). In this study, we use an Ellipsoid uncertainty set U , which is parametrized by a mapping matrix Δ and the Ellipsoid radius Ω . The mapping matrix Δ is derived from the uncertain parameters' covariance matrix and captures the correlations between the different uncertain parameters. If Σ is the covariance matrix, Δ is obtained by the Cholesky decomposition as: $\Sigma = \Delta \Delta^T$. The uncertainty set of an uncertain parameter b is presented in Eq. (28).

$$U \in \{b : \bar{b} + \Delta \xi, \|\xi\|_2 \leq \Omega\} \quad (28)$$

where $\bar{b}$ is the nominal value of the uncertain parameter, Δ is the mapping matrix and Ω is defined by the user to control the level of robustness. By maximizing the uncertain parameters, we obtain the robust counterpart (RC) problem, which is a deterministic equivalent for the RO problem. It is noted that the maximization is done per constraint (row) independently. To facilitate the row-wise representation of the RC in matrix notation, we introduce the standard basis vector $e_j \in \mathbb{R}^m$, where m is the number of rows in the constraints matrix, such that the j^{th} entry of e_j equals 1 and all other entries are 0.

$$\begin{aligned} \min_{x_I} f \\ \text{s.t.:} \\ \max_{\xi \in U} e_j^T [B(k_1 b(\xi) + k_2 x_I) - c] \leq 0 \quad \forall j \end{aligned} \quad (29)$$

Ben-Tal and Nemirovski (1999) shows that the solution of the RC problem defined by Eqs. (28) and (29) is given in Eq. (30).

$$\begin{aligned} \min_{x_I} f \\ \text{s.t.:} \\ e_j^T [B(k_1 \bar{b} + k_2 x_I) - c] + \Omega \|e_j^T Bk_1 \Delta\| \leq 0 \quad \forall j \end{aligned} \quad (30)$$

While Eq. (30) suggests a robust tractable formulation for the

integrated PDS-WDS operation problem, it requires that all the decisions are determined in advance of the optimized period, not utilizing uncovered uncertainty during the policy implementation. For that purpose, we formulate the adjustable version of RO, which is the ARO.

2.3. Adjustable robust optimization

Converting the static RO to its adjustable form requires converting the independent decision variables into linear decision rules (LDR) such that the decision vector x is parameterized, as given in Eq. (31).

$$x_I(\xi) = \pi_0 + \pi_1 b(\xi) \quad (31)$$

where π_0 represents the fixed component of the policy, and $\pi_1 b(\xi)$ defines the feedback term that adjusts decisions in response to realized uncertainty. π_0 is a vector of the same length as x_I and π_1 is a matrix with rows as the length of x_I , and columns equal to the length of $b(\xi)$.

Next, as in RO, the RC is obtained by maximizing the LHS of each row w.r.t the uncertain variables over the space of uncertainty to obtain the constraint-wise worst-case scenario. We note that in this formulation, the LDR parameters π_0 and π_1 become the new decision variables in the problem after substituting Eq. (31) in Eq. (29), which results in Eq. (32).

$$\begin{aligned} \min_{\pi_0, \pi_1} f \\ \text{s.t.:} \\ \max_{\xi \in U} e_j^T [B(k_1 b(\xi) + k_2 [\pi_0 + \pi_1 b(\xi)]) - c] \leq 0 \quad \forall j \end{aligned} \quad (32)$$

To solve the maximization problem in Eq. (32), we want to separate the deterministic terms from the uncertain ones. For ease of notation, we introduce z_0, z_1 such that

$z_0 = Bk_2 \pi_0 - c$, and $z_1 = Bk_1 + Bk_2 \pi_1$. Thus, the problem can be written as in Eq. (33), where a solution to the inner maximization problem is given by Ben-Tal and Nemirovski (1999).

$$\begin{aligned} \min_{\pi_0, \pi_1} f \\ \text{s.t.:} \\ \max_{\xi \in U} e_j^T [z_0 + z_1 b(\xi)] \leq 0 \quad \forall j \end{aligned} \quad (33)$$

Similarly to the RO, we end up with the RC form of the problem, which after replacing the decision variables with LDR, is referred to as the affine adjustable robust counterpart (AARC) and presented in Eq. (34).

$$\begin{aligned} \min_{\pi_0, \pi_1} f \\ \text{s.t.:} \\ e_j^T [z_0 + z_1 \bar{b}] + \Omega \|e_j^T z_1 \Delta\| \leq 0 \quad \forall j \end{aligned} \quad (34)$$

Noteworthy is that the RO can be considered a private case of the ARO by constraining all elements of the matrix π_1 to zero. The final version presented in Eq. (34) is a standard second-order conic optimization problem that can be solved with commercial solvers; hence, we end up with a tractable formulation for the adjustable optimal operation of integrated PDS and WDS.

2.4. Non-anticipativity

In multi-stage optimization problems under uncertainty, non-anticipativity is a crucial modeling principle that ensures decisions at each time step are made based solely on information that has been revealed up to that point. This prevents the optimization from exploiting knowledge of future uncertainties, thereby guaranteeing that the resulting decision policies are both implementable and realistic in practice. In this context, forecasts are known inputs at the time of

decision-making, while their future deviations from reality remain uncertain. The non-anticipativity constraint ensures that the model cannot rely on those future deviations before they are realized. It is noted that the adjustability of the formulation does not depend on whether the uncertainty itself changes or propagates over time. The adaptivity arises from the sequential process of observing uncovered uncertainty and updating decisions accordingly, which is enforced by the non-anticipative structure of the decision rules. To force non-anticipativity of the ARO decision rules, part of the matrix π_1 should be nullified (Jamal et al., 2019, 2018) such that the decisions at time t depend only on revealed uncertainty up to time $t-1$.

In general, these non-anticipative constraints could be written as: $g(\pi_1) = 0$, where g is an operator that decodes the non-anticipative pattern of π_1 . To ensure the LDR relies only on information from previous time steps and prohibits any reliance on future realizations, a subset of π_1 entries is constrained to be equal to 0, as shown in Eq. (35), which is added to the optimization problem in Eq. (34).

$$g(\pi_1) = \pi_1 \odot (1 - \Lambda) = 0 \quad (35)$$

where $\odot$ is element-wise (Hadamard) multiplication, and Λ is a binary non-anticipative matrix where the 0 and 1 values dictate which elements of π_1 will be constrained to be 0.

To illustrate this concept, consider a decision variable x that denotes the flow of a pump in an optimization problem consisting of 3 time steps. In dynamic decision-making problems, x should be adjusted based on uncovered uncertainties (in our case, from both the power and the water systems, ξ_t^P , ξ_t^W , respectively). As such, the corresponding LDR of x will be as in Eq. (36).

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} \pi_{0,1} \\ \pi_{0,2} \\ \pi_{0,3} \end{bmatrix} + \begin{bmatrix} \pi_{1,1,P_1} & \pi_{1,1,P_2} & \pi_{1,1,P_3} & \pi_{1,1,W_1} & \pi_{1,1,W_2} & \pi_{1,1,W_3} \\ \pi_{1,2,P_1} & \pi_{1,2,P_2} & \pi_{1,2,P_3} & \pi_{1,2,W_1} & \pi_{1,2,W_2} & \pi_{1,2,W_3} \\ \pi_{1,3,P_1} & \pi_{1,3,P_2} & \pi_{1,3,P_3} & \pi_{1,3,W_1} & \pi_{1,3,W_2} & \pi_{1,3,W_3} \end{bmatrix} \begin{bmatrix} \xi_1^P \\ \xi_2^P \\ \xi_3^P \\ \xi_1^W \\ \xi_2^W \\ \xi_3^W \end{bmatrix} \quad (36)$$

Since the first decision at time 1 has no uncovered uncertainty, all the dependencies on ξ_t^P, ξ_t^W should be nullified. Thus, the first row of the matrix π_1 should be defined as a row of zeros. In the second time step, the uncertainty of the time step is already uncovered, thus the x_2 can depend on ξ_1^P and ξ_1^W . The other coefficients in the second row should be nullified to enforce non-anticipativity. In general, the non-anticipative matrix is constructed such that the elements of π_1 that are multiplied by the uncertainty of the later time steps will be 0. Thereby, a decision variable represents a decision at time t will be a function of the uncovered uncertainty up to time $t-1$: $x_t = f(\xi_1^P, \dots, \xi_{t-1}^P, \xi_1^W, \dots, \xi_{t-1}^W)$. Using Eq. (35) and defining Λ as in Eq. (37) guarantees the non-anticipativity conditions of the optimal policy. Without these conditions, the optimal policy will be non-realistic, as it could exploit information about future uncertainties that would not be available in practice at the time of decision-making.

$$\Lambda = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 \end{bmatrix} \quad (37)$$

In fact, our modeling approach that defines the pattern of dependency of the decisions with respect to the uncertainty allows for further complex modeling of the available information beyond non-anticipativity. The matrix presented in Eq. (37) can be modified to capture patterns like data latency or limited data lags by setting more elements of the matrix to zero. Moreover, different information availabilities can be assigned to each of the systems. For example, the LDRs of

the water decisions can either depend on the uncovered uncertainty of the water system alone or on both the power and the water systems. Similarly, the power LDRs can consider only one of the systems or both of them. Such extensions of the information limitation constraint can be done easily by declaring similar Λ matrices that nullify more elements of π_1 . Then add the additional matrices to the multiplication defining the operator g in Eq. (35). It is expected that performance under limited information conditions will be inferior compared to the full information availability, affected only by the non-anticipativity.

3. Results

The method is demonstrated through two case studies aiming to optimize the integrated day-ahead operation of PDS-WDS for the day-ahead optimal dispatch. We focus on case studies that represent national-level systems, where water facilities such as desalination plants and major pumping stations impose significant loads on the power grid, making coordination between the two systems essential for robust and cost-effective operation (Salomons et al., 2023). The case studies are designed to represent the aggregated main core of national-scale power and water systems. At this level, each bus corresponds to a major transmission load, and each water demand represents the entry to a local distribution subsystem. The intention of these case studies is therefore not to model detailed hydraulics or nodal voltages, but to capture the system-level interaction and scheduling decisions that occur on a day-ahead planning horizon. Under these settings, aggregated pumping, desalination, and storage decisions dominate the overall energy-water nexus. Therefore, despite their compact schematic representation, the networks used in this study should not be interpreted as small or simplified systems, but as macro-scale abstractions of large national infrastructures, which is the appropriate resolution for day-ahead operational coordination.

The uncertainty for both case studies is defined by a covariance matrix that captures the standard deviation of each parameter and the correlations between the different uncertain parameters. It is assumed that the parameters are only temporally correlated w.r.t previous time steps and no correlation exists between the different loads, water consumers, and the PV radiation. Recalling that the uncertain variables vector is defined as $b(\xi) = \bar{b} + \Delta\xi$, the nominal values and the covariance matrix are sufficient information to build the uncertainty set.

For each case study, a series of experiments was conducted under varying levels of robustness, controlled by the uncertainty set radius Ω . As Ω increases the uncertainty set captures a broader range of potential deviations from the nominal scenario, hence producing solutions that are robust to increasingly extreme conditions. Each experiment consists of two stages: an optimization stage in which RO and ARO are solved for a given value of Ω , and a simulation stage where the obtained solution is tested against a set of 1,000 random scenarios drawn from a normal distribution with the same statistical moments as those that define the uncertainty set. Based on this set of scenarios, the performance is evaluated in terms of optimality and robustness.

Notably, the nature of the two methods' solutions is different. ARO produces a scenario-dependent decision rule, allowing decisions to adapt to realized uncertainty. Accordingly, its performance is computed based on the simulated random scenarios. In contrast, the decision obtained from RO is usually static and independent of the realized scenario. However, due to the equality elimination discussed above, the dependent decision variables $x_{\mathcal{D}}$ act as recourse decision variables, thus the RO solution exhibits scenario dependence with regard to the decisions $x_{\mathcal{D}}$, while the independent decision variables x_I are fixed as presented in Eq. (25).

To evaluate robustness across both approaches, we compute the Constraint Satisfaction Rate (CSR), defined as the proportion of scenarios in which all constraints are satisfied. This metric captures how

well a solution generalizes unseen realizations of uncertainty.

In addition, we report the Price of Robustness (POR), which quantifies the additional cost incurred by selecting a robust solution over a nominal (uncertainty-ignored) solution. This allows us to examine the tradeoff between robustness and cost efficiency as Ω increases.

3.1. Case study I

The first case study, presented in Fig. 1, serves as an illustrative example for validating the methodology. It includes a 3-bus PDS equipped with two diesel generators, a PV source, and a battery. The PDS supplies the power to a desalination plant, which serves as the source to a WDS composed of two tanks. A pumping station with three parallel pumps, also powered by the PDS, delivers water to a higher consumption zone.

The results of the RO and ARO experiments are summarized in [Table 1](#). Aligned with RO theory, increasing the uncertainty set size (Ω) leads to more conservative (more robust) but also more expansive solutions. Across all tested levels of robustness, the ARO produced lower cost policies compared to RO. However, the CSRs yielded by the two methods are very similar, with no obvious superiority to one of them. To better review the tradeoff between the two objectives, they should be analyzed together in a multi-objective space, as shown in [Fig. 2](#). The POR analysis shows that the ARO costs are consistently lower than those achieved by the RO for the same levels of robustness. This implies the overall superiority of ARO, where for a similar increase in robustness, the ARO pays less in optimality.

Fig. 2 describes the Pareto tradeoff between Cost and CSR for the RO (top) and ARO (bottom). Since the two policies depend on the random scenario realizations, in each experiment, we get a set of optimal costs and not a single value. The colored curves present the average cost across all scenarios and the CSR computed based on the same set of scenarios. Each point on the Cost-CSR curves corresponds to a different level of robustness (Ellipsoid radius), where lower values of Ω (bottom-left points) yield less conservative solutions. In contrast, higher Ω values (top-right points) yield more robust and costlier solutions. Superimposed on the curves are the distributions of actual costs obtained under the simulated scenarios. The histograms illustrate the variability inherent in each method and how it depends on the random scenarios. It can be seen that the realized costs are centered around the mean cost and resemble approximately a normal distribution, consistent with the distribution of the random scenarios. ARO solutions are shifted left (less costly) compared to RO, with the shift more significant in the higher levels of robustness (e.g., the more expansive, more robust solutions). The meaning is that ARO dominates the objective space at lower operational costs for the same levels of constraint satisfaction. This underscores the conservatism of RO and the advantage of the adjustable policies in efficiently managing uncertainty without incurring excessive cost.

Examining the decision space emphasizes the differences between the methods. Figs. 3–5 present the system's operation in response to

random scenarios under the RO and ARO policies. For better visibility, only 100 scenarios are plotted in Fig. 4. Fig. 3 shows the battery SOC trajectory. The blue lines represent trajectories under the ARO policy, while the red lines represent trajectories under the RO policy. In the case of RO, all the trajectories overlap to a single trajectory, which corresponds to the battery SOC across all scenarios. The RO formulation is responsive only to scenarios involving dependent variables because of the elimination strategy introduced in this paper. Since battery charging is chosen as an independent variable, the battery state is scenario-invariant. It can be seen that the system utilizes only a small portion of the available battery capacity, primarily operating within the range of approximately 200 MWh to 380 MWh, against a total capacity limit of 0-500 MWh. The SOC trajectories still exhibit a pronounced diurnal charging pattern, indicating the need to strategically manage power throughout the day. The battery completes two main charging cycles. The first is in the early morning, corresponding to off-peak consumption, followed by a significant rise throughout the daytime. This latter charging phase is carefully managed to utilize PV injections and ensure the battery reaches its maximum charge level just before the evening peak load (18:00-21:00).

Fig. 4 presents the generators' dispatch along the optimization horizon. Each subplot presents a single generator, where blue lines represent the generator dispatch results from the ARO policy across random scenarios, and red lines represent the same for the RO policy. The generator at bus 3 is a dependent variable and therefore absorbs most of the uncertainty in both methods. This figure illustrates the difference between dependent and independent variables. The left plot presents the dispatch of the generator at bus 2, which is an independent variable, while the right plot presents the generator at bus 3, which is a dependent variable. Here, we can clearly notice that in the RO policy, only dependent variables respond to scenario realizations. Hence, in the RO solution, only generator 3 shows a different dispatch trajectory for different scenarios. Generator 3 shows similar behavior in the two methods, where the realization dictates the generation. We note that the RO solution is based on "Here and Now" decisions made in advance, while the ARO trajectory is developed through implementation, allowing for recourse decisions that result in lower total expenses. Furthermore, in this case study, the two generators are identical, and the generated power is split approximately equally between them. Generator 2 maintains a more stable operation while generator 3, the dependent variable, absorbs most of the uncertainty; hence, we observe fluctuations in the generation trajectories.

In both methods, most of the uncertainty is absorbed by the water tanks. Similar to Figs. 3–5 presents the volume trajectories of the two methods across the scenarios. The ARO solution exhibits less dispersion compared to RO, due to its adaptive nature, that allows for dynamic adjustment of the decisions. In both tanks, the most restrictive constraint is the requirement that the volume at the end of the day be at least as high as the initial volume. ARO's adaptability enables a final volume much closer to the initial volume constraint. This means not only fewer

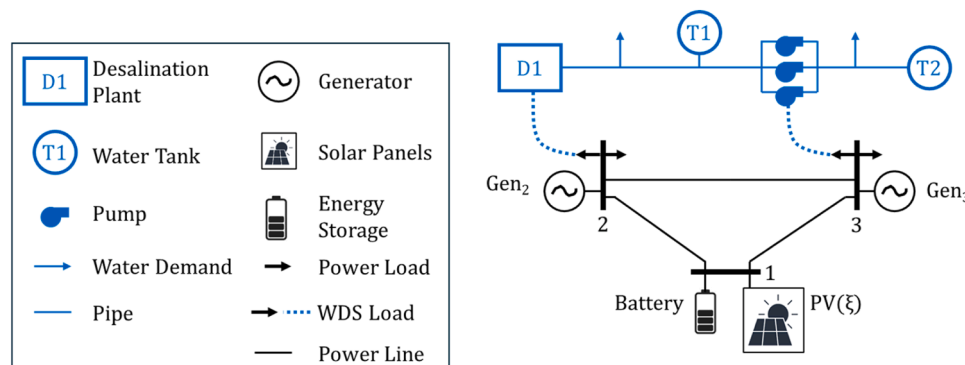
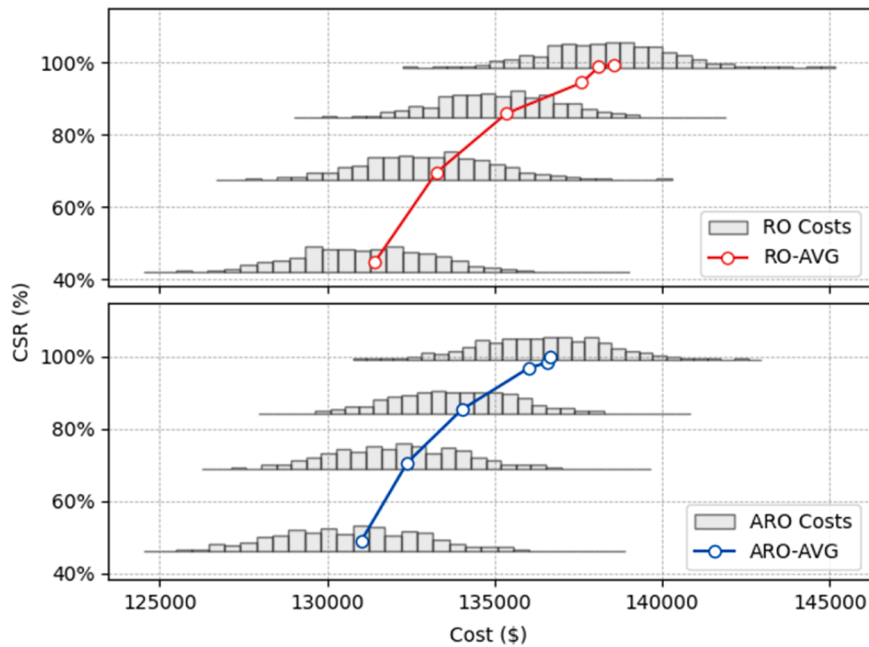
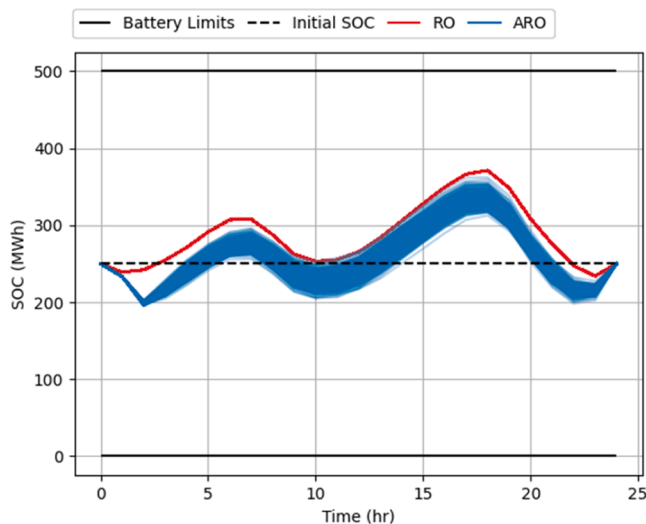


Fig. 1. Case study I layout.

Table 1

Case study I results.

Robustness Ω	RO				ARO			
	Average Cost (\$)	STD Cost (\$)	CSR (%)	POR (%)	Average Cost (\$)	STD Cost (\$)	CSR (%)	POR (%)
0	129,986	2,268	23.80%	-	129,986	2,268	23.80%	-
0.5	131,430	2,077	45.0%	1.11%	131,008	2,204	48.90%	0.79%
1	133,257	1,926	69.60%	2.52%	132,390	2,042	70.70%	1.85%
1.5	135,355	1,830	86.0%	4.13%	134,020	1,939	85.40%	3.10%
2	137,605	1,778	94.50%	5.86%	136,000	1,879	96.80%	4.63%
2.5	138,112	1,781	98.90%	6.25%	136,562	1,871	98.40%	5.06%
3	138,539	1,800	99.40%	6.58%	136,647	1,874	99.90%	5.12%

**Fig. 2.** Price of robustness tradeoff for case study I.**Fig. 3.** Battery SOC trajectory for case study I. Battery charging is an independent variable.

constraint violations (where final volume is below the initial one), but also fewer surpluses (where the final volume is above the initial one). Such surpluses indicate that water was pumped as a safety margin to meet this constraint, even though it was not ultimately needed, resulting in a suboptimal solution.

3.2. Case study II

The second case study is based on the IEEE-14 benchmark, as shown in Fig. 6. The benchmark PDS was modified to include an uncertain renewable source and a battery. In addition to meeting its conventional loads, the PDS also supplies the energy to a WDS with two desalination plants, five pumping stations, and five tanks. In this case study, the ARO lag window was reduced to 8 h due to hardware memory issues. This means the policy is adjusted based on the recent 8 h. Further details on this limitation are in the Computational Analysis section below. Uncertainty is introduced through variations in power loads, water demands, and renewable power generation, all statistically independent of one another. However, each uncertain parameter exhibits temporal correlation across time steps. This means that while the uncertainties are uncorrelated across different types, the realization of a given parameter at one time step is correlated with its values at other time steps.

The typical tradeoff between optimality and robustness is also observed in this case study, as presented in Table 2 and Fig. 7. Here, the advantage of ARO is more pronounced, as the CSR of the ARO solution starts above 50%, whereas the CSR of the RO is below 20%. CSR in ARO increases more rapidly with cost, indicating a greater adaptability to uncertainty. Nevertheless, the ARO curve is consistently shifted to the left, showing that higher reliability is achieved at lower operating costs. The STD in both methods decreases with larger uncertainty sets (larger Ω values). As Ω increases, the solutions become more conservative, leading to reduced sensitivity to scenario variations. This reduced sensitivity narrows the spread of objective values across scenarios and

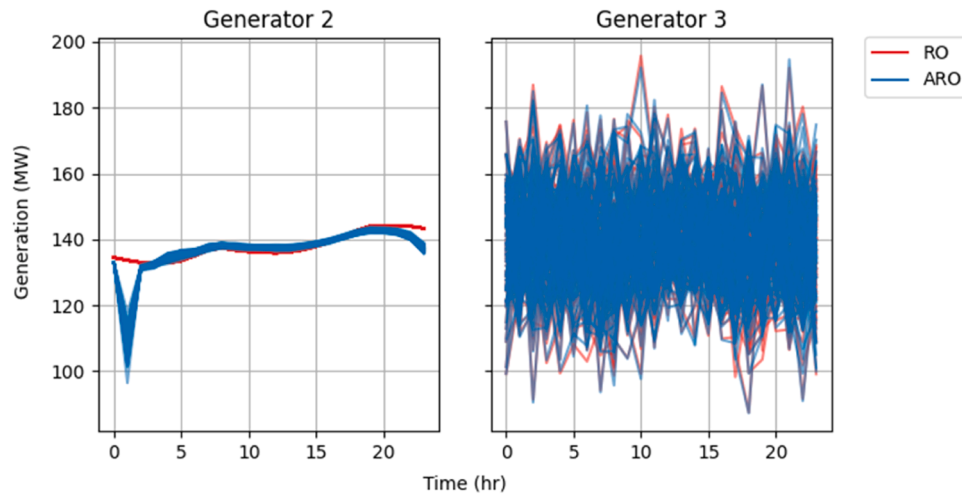


Fig. 4. Generators dispatch for case study I. The left plot represents Generator 2 (independent variable), and the right plot represents Generator 3 (dependent variable).

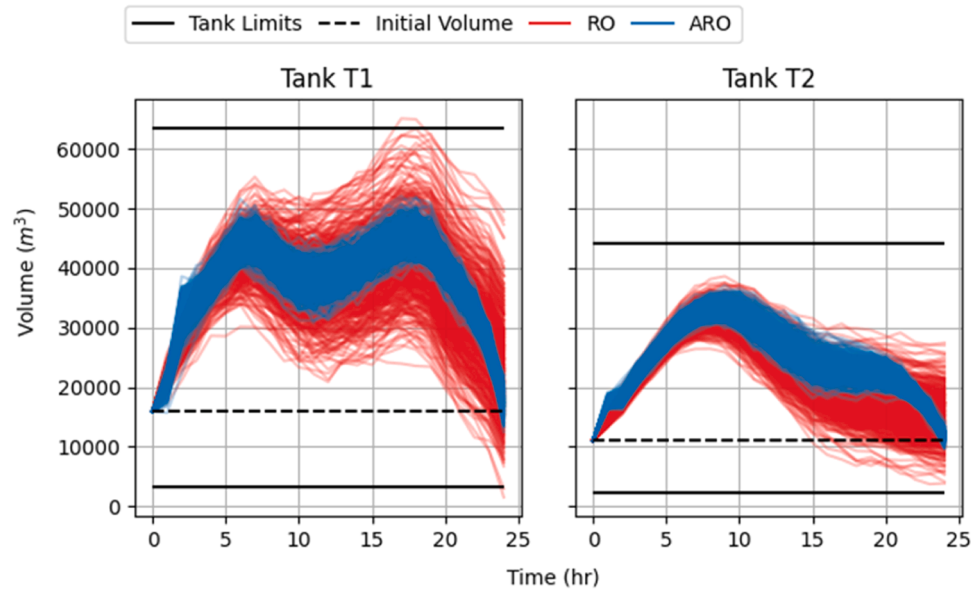


Fig. 5. Tank volume trajectories for case study I. Tank inflows are dependent variables.

therefore results in lower STD. Nonetheless, the STD of the ARO method is slightly higher than that of the RO, indicating that the ARO dependency in the scenarios is higher, since the RO depends on the scenario realizations only due to the elimination of the equality constraints.

The tradeoff between optimality and robustness is presented in Fig. 7, with the same structure as in Fig. 2. The RO method shows low robustness, with a CSR range of 20%–40%, while the ARO starts with above 50%. Constraint satisfaction increases steeply with increasing robustness values for ARO, while for RO, the increase is more gradual. Furthermore, the ARO increase in cost is also more moderate compared to RO. This illustrates the ARO superiority, with the ARO curve overall shifted to the top left, meaning lower costs and higher CSR than the RO solutions. The results of the operational dynamics for this case study are attached as supplementary material, S1.

3.3. Reliability metrics analysis

To further analyze the methods' performance, Fig. 8 compares several additional reliability metrics computed across the same random

scenarios. Each line represents a single scenario evaluated against optimized policies (RO or ARO). The metrics include the total operation cost, average and maximum ramping of the generators, the total violated generators dispatch (sum of power above or below the generator bounds), water volume (sum of tank shortages and overflows), and the variability of pumping operation expressed as the average standard deviation of flows across the pumping stations. The figure highlights that ARO solutions are better for operational costs, while exhibiting slightly higher generator ramping. This reflects its adaptive nature, which allows generators to adjust dynamically to realized conditions. ARO yields more frequent redispatch actions. The RO solutions exhibit a greater number of violations. It can be seen that all violations occur in the tank volumes, where the generators' bounds are not violated. The last metric of pumping stations flow variability is the only one where RO performs better than ARO. As explained above, the pumping variables were chosen to be independent for the decomposition method; therefore, in RO, they are not influenced by scenario realizations and demonstrate constant

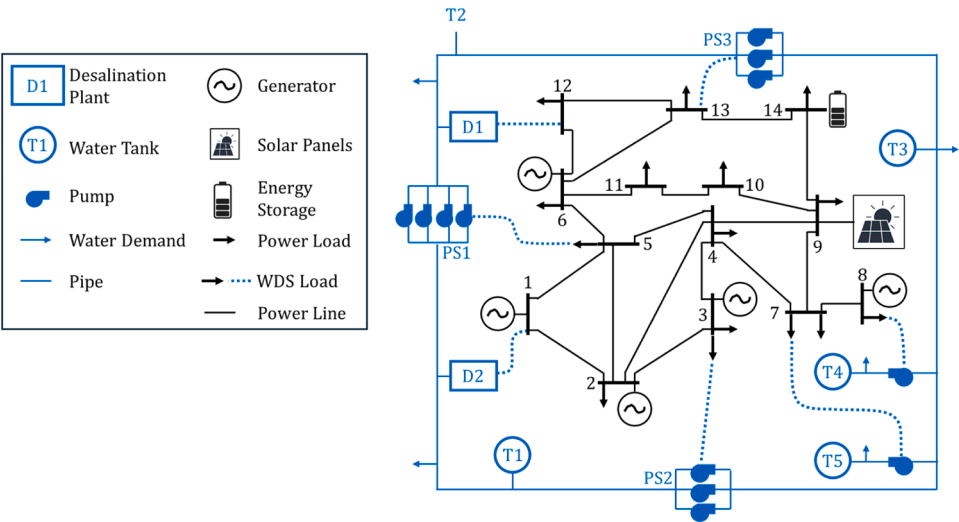


Fig. 6. Case study II layout.

Table 2
Case study II results.

Robustness Ω	RO				ARO			
	Average Cost (\$)	STD Cost (\$)	CSR (%)	POR (%)	Average Cost (\$)	STD Cost (\$)	CSR (%)	POR (%)
0	371,965	1,998	3.20%	-	371,965	1,998	3.20%	-
0.5	373,653	1,908	14.40%	0.45%	374,449	1,977	52.40%	0.67%
1	375,512	1,836	42.20%	0.95%	375,401	1,945	78.10%	0.92%
1.5	377,500	1,788	69.70%	1.49%	376,419	1,927	92.10%	1.20%
2	379,639	1,763	88.70%	2.06%	377,466	1,919	97.60%	1.48%
2.5	381,807	1,755	96.20%	2.65%	378,526	1,917	99.80%	1.76%
3	383,987	1,752	99.10%	3.23%	379,596	1,915	100.00%	2.05%

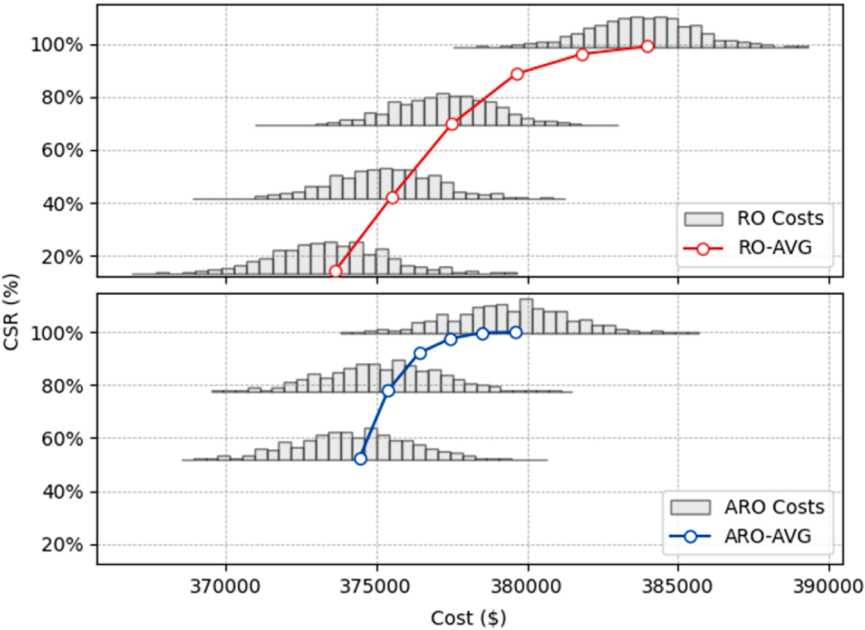


Fig. 7. Price of robustness tradeoff for case study II.

3.4. Adaptability analysis

Fig. 9 illustrates the adaptability of the ARO policy when exposed to two distinct realizations of uncertainty, using the same optimized policy. In the first scenario (blue lines), the system faces a day with water demands above nominal levels. As expected, the water facilities of

desalination plants and pumping stations adjust their operation starting from the early hours, producing and conveying larger volumes of water. Consequently, tank storage levels are changed accordingly. This increased water activity leads to higher electricity consumption, prompting Generator 1 to increase its dispatch to supply the additional power required for pumping and desalination. Other generators, whose

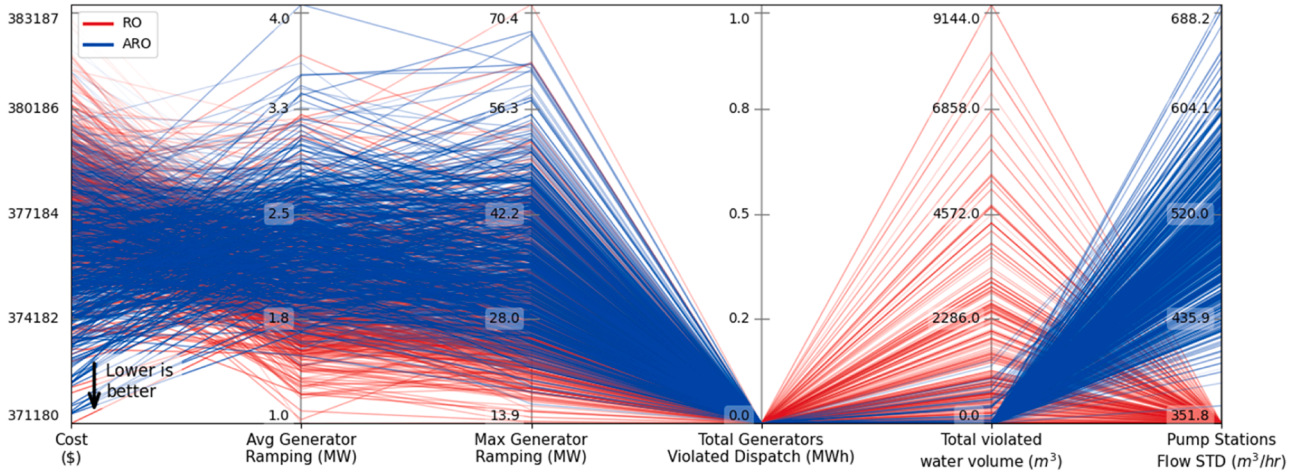


Fig. 8. RO and ARO reliability metrics across random scenarios.

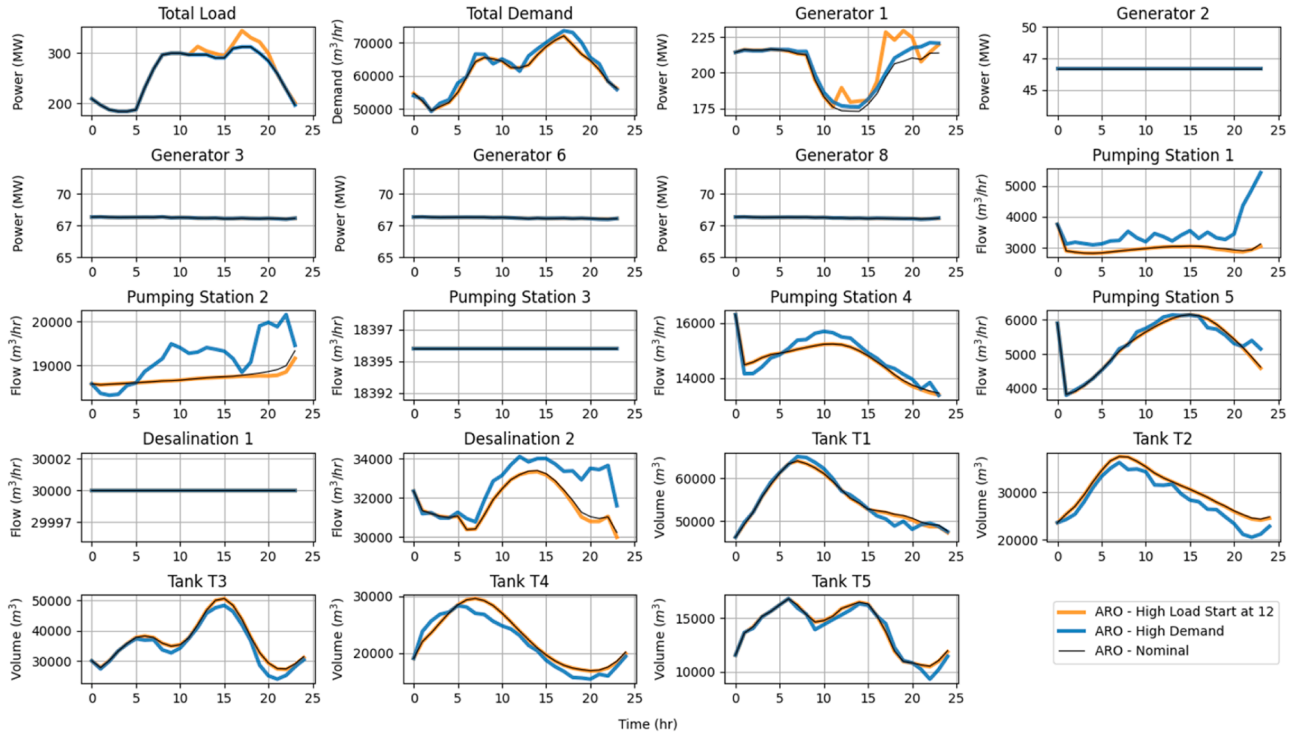


Fig. 9. Adaptability of the ARO policy under two uncertainty realizations: high water demand (blue) and midday power load surge (orange).

operation is unaffected by the uncertain parameters, maintain identical, constant dispatch levels. It is noted that Generator 1 is a dependent variable in the presented constraint elimination methodology, while all other generators are independent.

The second scenario (orange lines) represents a day when an unexpected surge in power load begins at midday (12:00). The ARO policy dynamically responds to this new condition by increasing the dispatch of Generator 1 to meet the additional electricity demand. Interestingly, the water system exhibits only marginal adjustments, mainly a slight reduction in operation at Pumping Station 2 and Desalination Plant 2, observed toward the end of the day. These responses occur only after the load surge begins, demonstrating the decision rule's real-time adaptability.

Overall, the results reveal that while water demand variations strongly influence the power system operation, the reverse effect is more limited. This asymmetry is physically intuitive: increases in water demand directly raise electricity consumption for pumping and

desalination, whereas power load variations mainly influence the timing of water operations without changing the total desalinated or pumped volume. The observed behavior confirms that the ARO policy provides an operationally meaningful adaptive response to evolving uncertainties while maintaining stability and feasibility across both infrastructures.

3.5. Data latency

Following the discussion above on non-anticipativity and data availability issues, a sensitivity analysis was conducted to examine the impact of delayed information on adaptive decision-making. Data latency can arise when the online data used to update the optimal solution in ARO is received with a delay. Such a delay can be due to sensor sampling and transmitting frequency or to data integration processes. In the case of integrated systems, the risk of data latency or unsynchronized data increases as data from different systems must be processed and sent to a central control room. The analysis examined various latency

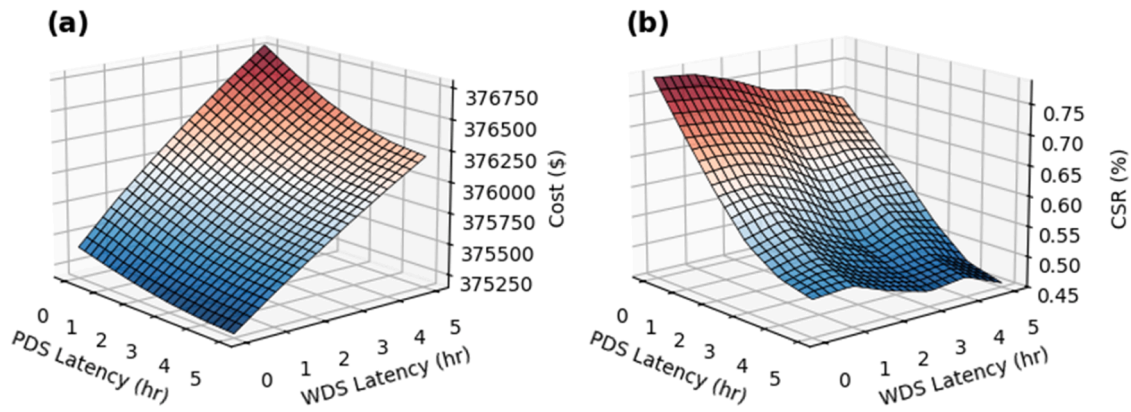


Fig. 10. Effect of data latency on ARO performance: (a) average operation cost, (b) constraint satisfaction rate (CSR).

durations in the PDS and WDS.

Fig. 10 illustrates the sensitivity of the ARO policy to data. Latency is implemented by modifying the non-anticipativity matrix, thereby limiting the uncertainty information accessible to the adaptive decision rules. The two surfaces show the latency effect observed separately in the PDS and WDS. The performance degradation is evaluated by the average total operation cost (a) and the (CSR) (b), over the same random scenarios used for the previous analysis.

The impact of WDS latency shows that greater latency reduces the information basis, consistently raising costs and reducing CSR. In contrast, PDS latency exhibits a counterintuitive pattern. As latency increases, the average cost decreases while the CSR drops sharply. This indicates that delayed adaptation on the power side effectively relaxes the robustness of the solution, yielding cheaper but less reliable operation. The steeper CSR slope along the PDS latency axis confirms this tradeoff. Overall, these results demonstrate that data latency affects the two systems asymmetrically. Timely WDS data are crucial for cost efficiency, while timely PDS data are critical for maintaining reliability.

3.6. Computational analysis

All computational experiments were performed on a personal laptop equipped with an Intel Core i7-12700K CPU (2.3 GHz), 64 GB SO-DIMM RAM, running Windows 11 (64-bit). The optimization models were implemented in Python 3.10 using the CVXPY 1.7.0 modeling package and solved with the MOSEK 11.0.29 commercial solver. The RO and ARO formulations yield linear programs (LP) and second-order cone programs (SOCP), respectively. MOSEK solves these using specified variations of the interior-point algorithm. Table 3 summarizes the computational properties and run times of both case studies and two additional case studies, whose results are attached as supplementary materials (S2 and S3). The table reports the number of ARO lags, the number of decision variables, the number of second-order cone (SOC) constraints, the canonical matrix dimensions, the problem sparsity, and the solver run times reported based on the average of the same

experiments presented in the results section. It should be noted that the number of decision variables and SOC constraints seems identical between the RO and ARO formulations. This results from a unified implementation approach in which the RO model internally calls the ARO formulation while constraining all adjustable components to zero. For the RO formulation, these dimensions therefore reflect the high-level modeling interface rather than the actual problem passed to the solver. Once the adjustable components are fixed to zero, the solver's presolve routine removes them, drops the conic structure they induce, and collapses the problem into a pure LP. Consequently, the RO instances solve dramatically faster, despite showing the same dimensions at the modeling stage.

The SOC constraints are converted into a canonical conic form by introducing auxiliary variables and expanded matrix blocks. This canonicalization can significantly inflate the effective problem size seen by the solver. In our ARO formulation, each decision variable is represented as a vector of coefficients (the "lags window") multiplied by the uncertain parameters. Consequently, every adjustable variable is converted to a vector of variables. The combination of these two techniques explains the larger canonical matrices observed for ARO.

During this study, a memory limitation was encountered for the largest instances.

While the mathematical dimensions were moderate, the fully adjustable ARO combined with ellipsoidal uncertainty sets required converting every affected constraint into SOC form.

The issue stems from memory overhead, not algorithmic intractability. The problems remain convex and solvable but exceed the RAM capacity of a personal laptop. To preserve tractability, we adopted a reduced ARO lag window, reported in Table 3, which limits the number of adjustable coefficients while retaining the most recent historical information. The reduction is conceptually justified, as recent lags carry the most relevant information for operational decision-making. In large-scale or production implementations, this limitation can be mitigated by using low-level languages (e.g., C++) or high-performance computing environments that manage sparse data structures more efficiently than

Table 3
Computational performance.

Case Study	Method	ARO Lags	Vars	SOC Constr.	Solver Matrix Rows x Cols ($\times 10^3$)	Solver Matrix Non-Zeros ($\times 10^6$)	Solver Matrix Density	Solution Time Average (s)
Illustrative	RO		20,953	888	11.0 $\times$ 129.2	0.06	0.004%	0.1
Illustrative	ARO	24	20,953	888	11.0 $\times$ 108.5	27.41	2.289%	104.5
IEEE-9	RO		97,273	1,824	48.7 $\times$ 713.6	0.23	0.001%	0.7
IEEE-9	ARO	24	97,273	1,824	48.7 $\times$ 616.9	155.71	0.518%	1155.5
IEEE-14	RO		231,001	2,136	113.1 $\times$ 1209.1	0.30	0.0002%	1.0
IEEE-14	ARO	8	231,001	2,136	113.1 $\times$ 978.7	327.34	0.296%	262.5
IEEE-24	RO		444,361	3,096	216.5 $\times$ 2605.2	0.45	0.0001%	2.3
IEEE-24	ARO	3	444,361	3,096	216.5 $\times$ 2161.7	564.77	0.121%	4045.8

Vars – Decision Variables, Constr. – Constraints, Cols – Columns

high-level modeling interfaces such as the CVXPY modeling package.

While the present analysis focuses on day-ahead scheduling, the same RO and ARO formulations, along with the equality constraints elimination method, are fully compatible with rolling-horizon implementations, in which the uncertainty set and decision variables are updated as new information becomes available. Such sequential re-optimization would preserve the same tractable conic structure and can benefit from warm-starting or parallelization to further reduce computation time. The RO models are solved in a few seconds to minutes, making them fully compatible with rolling horizon frameworks. The ARO models require longer solution times, ranging from several minutes to over an hour for the largest case. Yet, it remains within a reasonable computational range, which can be further reduced as detailed above.

4. Discussion

The proposed method, which reparametrizes the decision vector by decomposing it into independent and dependent variables, provides a key step to implement RO in problems involving uncertain equality constraints. By reformulating these constraints and eliminating the dependent variables, the method allows a tractable robust counterpart formulation, benefiting from the advantages of the RO paradigm. At the same time, a critical question arises about how to select the independent variables. As discussed in the methodology section, pure mathematical approaches such as minimal spanning trees, reduced row Echelon form (RREF), QR decomposition, and Null space analysis can be taken to identify a valid decomposition. However, in physical systems, the selection of independent variables has physical implications. Different decompositions lead to different forms of inequality constraints, Eq. (25), which in turn determine how uncertainty is propagated through the constraints. A decomposition that amplifies the uncertainty can lead to overly conservative constraints and even infeasible solutions.

To illustrate this, consider a simplified PDS with two generators and a single load, as presented in Fig. 11, and suppose that generator G_1 has lower dispatch costs. Depending on the decomposition, one generator will be independent, and the second generator can be expressed as the difference between the load and the first generator. If G_2 is dependent, it must adjust to match any load deviations. In this setup, G_1 can operate closer to its economic optimum. However, if extreme peak loads are expected and G_2 has tighter capacity bounds, this setup may result in constraint violations. Alternatively, setting G_1 as the dependent generator allows it to flexibly absorb uncertainty at the expense of deviating from its cost-efficient dispatch. Thus, the choice of the dependent variable influences both the economic performance and robustness of the solution.

In some cases, the dependent element is more intuitive. For example, in WDS, tank volumes can be considered state variables that depend on the pumping variables and the demand realizations. While batteries in PDS function similarly to water tanks, their economic model is different and dictates explicit decision-making of charging and discharging. This means that the modeler needs to choose the dependent variable that will absorb the unknown fluctuations. In case a generator is dependent, it will adjust its output to match load variations. If a battery is dependent, its charging and discharging will be adjusted to maintain the power balance. This choice will impact the objective value as well as constraint

violations. If a generator is dependent, the generation can violate its lower and upper bounds. If a battery is dependent, the SOC can violate the min and max capacities. In this study, we conducted several trials and chose the following configuration: all generators but the most efficient ones are independent, and battery charging is also independent, while the remaining generator and tank inflows are dependent. A set of results showing the dynamics of the system with a different configuration of the independent variables can be found in the supplementary materials, S4. This topic of independent and dependent variables decomposition and its impact on robust solutions is marked as highly interesting future research that can shed more light on how to improve robust optimization modeling and obtain superior solutions.

5. Conclusions

This paper presents RO and ARO methods for optimizing the day-ahead dispatch of integrated PDS-WDS under uncertainties to minimize the overall power generation cost. By coupling the two systems, we allow a dynamic response to consumers' behavior and PV scenarios rather than relying on static electricity tariffs that govern the WDS load shifting. Such a collaborative framework is particularly crucial given the increased penetration of intermittent renewable energy sources and the heightened variability in both power and water demand.

The proposed methods are particularly effective for such problems, as they avoid precise stochastic modeling and offer tractable formulations. The study presents an approach for formulating equality balance constraints based on dependent variables elimination, reducing the problem dimensions, and relaxing it to inequality constraints only. This approach is then implemented in two case studies demonstrating the tradeoff between optimality and robustness. Compared to deterministic optimization, which exhibits very low constraint satisfaction rates (23.8% and 3.2% for both case studies), the proposed RO and ARO increase robustness to above 99% with only a 2–5% increase in operational costs. This confirms that the proposed RO and ARO formulations deliver high resilience at minimal economic penalty. Additionally, the results highlight the superiority of ARO over RO, as ARO offers a more flexible solution that allows for adjustments based on real-time measured data. The ARO policies achieve consistently lower operating costs and exhibit fewer constraint violations compared to the RO policies. This improvement arises from the ability to dynamically react to unfolding uncertainties, such as fluctuating renewable generation or unexpected shifts in water demand.

Several sensitivity analyses were conducted to demonstrate the ARO's adjustability across different scenarios, revealing the impact of each system on the coupled one. Reliability was evaluated through various metrics and visualized across a range of random scenarios. The impact of data availability was examined, illustrating how latency in data measurements in each system impacts the operational costs and robustness. The decomposition of equality constraints into independent and dependent variables was found crucial and identified as a key avenue for future research.

CRedit authorship contribution statement

Gal Perelman: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mashor Housh:** Writing – review & editing, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Aviad Navon:** Writing – review & editing, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Avi Ostfeld:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

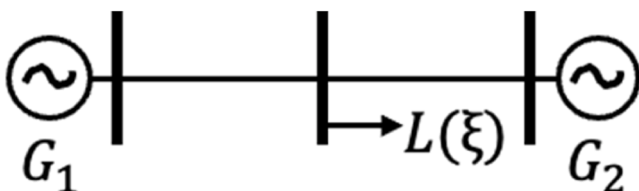


Fig. 11. Illustrative independent variables selection example.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Avi Ostfeld reports financial support was provided by The Israeli Water Authority. Mashor Housh reports financial support was provided by The Israeli Science Foundation. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work

reported in this paper.

Acknowledgments

This research was supported by the Bernard M. Gordon Center for Systems Engineering at the Technion under project number 2034792, the Israeli Water Authority under project number 2034739, and the Israeli Science Foundation under grant 2480/22.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.watres.2026.125325](https://doi.org/10.1016/j.watres.2026.125325).

Appendix A. Nomenclature Table

Notation	Meaning
$\mathcal{B}$	Set of all buses
$\mathcal{G}$	Set of all generators
$\mathcal{E}$	Set of all batteries
pc	Pump combination (a subset of pumps in a pumping station operated simultaneously)
$\mathcal{D}$	Set of all desalination plants
K	Number of piecewise linear segments
$\Theta_{\text{pc}}(i)$	Set of all pump combinations connected to bus i
$\Theta_{\mathcal{D}}(i)$	Set of all desalination plants connected to bus i
$\mathcal{S}$	Set of all storage tanks
n	Number of buses in the system
T	Number of time steps
t	Index of time steps
Δt	Time step size
I	Set of independent decision variables
O	Set of dependent decision variables
U	Uncertainty set
Ω	Ellipsoid radius
Δ	A mapping matrix that defines the uncertainty set; computed from a covariance matrix Σ
Λ	A matrix represents the non-anticipativity pattern
Optimization Decision Variables	
$\theta_{i,t}$	Voltage angle at bus i and time t
$w_{i,t}$	Auxiliary variables for the piecewise linear approximation of the cost function for generator i at time t
$x_{i,t}^g$	Active power generated at bus i and time t
$x_{i,t}^{\text{bat}}$	Battery charging of battery at bus i and time t (negative for discharging)
$x_{\text{pc},t}^p$	Portion of time step where the pump combination pc is activated at time t
$x_{d,t}^D$	Volume of desalinated water at desalination plant d and time t
$x_{s,t}^{\text{tank}}$	Flow into tank s at time t (negative for outflow)
$\text{SOC}_{i,t}$	State of charge of battery at bus i and time t
$V_{s,t}$	Volume of tank s at time t
π_0, π_1	LDR coefficients (i.e., decision variables) for the ARO formulation representing
ξ	Uncertain variable representing fluctuations from nominal values
Parameters	
a_i, b_i, c_i	Coefficients of the quadratic cost function of generator i
$\alpha_{i,k}, \beta_{i,k}$	Coefficients that define the k^{th} linear segment of the cost function of generator i
$L_{i,t}$	Active power load at bus i and time t
$PV_{i,t}$	Renewable power injection at bus i and time t
$\underline{\theta}, \bar{\theta}$	Min and max bounds of the voltage angle
$\underline{x}_i^g, \bar{x}_i^g$	Min and max bounds of the power generation at generator i
$\underline{x}_i^{\text{bat}}, \bar{x}_i^{\text{bat}}$	Min and max bounds of the battery charging/discharging rate at bus i
$\underline{x}_k^D, \bar{x}_k^D$	Min and max bounds of the desalination production at desalination plant k
$\underline{\text{SOC}}_i, \bar{\text{SOC}}_i$	Min and max bounds of the state of charge of the battery at bus i
$\underline{V}_s, \bar{V}_s$	Min and max bounds for water volume at tank s
$\text{SOC}_{i,\text{init}}, \text{SOC}_{i,\text{final}}$	Initial and final state of charge of the battery at bus i
$V_{s,\text{init}}, V_{s,\text{final}}$	Initial and final water volume at tank s
R_i	Max up/down ramp rate of generator i
$B_{i,j}$	Element of susceptance between bus i and bus j
q_{pc}	Nominal flow of pumps combination pc
P_{pc}	Nominal power of pumps combination pc
P_d	Power (per 1 m ³ of water) required for the desalination plant d
$d_{s,t}$	Water demand from tank s at time t

Data availability

The data and code used in this research are available on the following link: <https://github.com/GalPerelman/power-water-aro>.

References

- Alhazmi, M., Dehghanian, P., 2021. Optimal integration of interconnected water and electricity networks. *IET Gener. Transm. Distrib.* 15, 2033–2043. <https://doi.org/10.1049/GTD2.12153>.
- Alhazmi, M., Dehghanian, P., Nazemi, M., Oikonomou, K., 2023. Uncertainty-informed operation coordination in a water-energy nexus. *IEEE Trans. Industr. Inform.* 19, 6439–6449. <https://doi.org/10.1109/TII.2022.3195695>.
- Alnahhal, Z., Shaaban, M.F., Hamouda, M.A., Majazi, T., Bardan, M.A., 2023. A water-energy nexus approach for the co-optimization of electric and water systems. *IEEE Access* 11, 28762–28770. <https://doi.org/10.1109/ACCESS.2023.3257858>.
- Basak, S., Dey, B., Bhattacharyya, B., 2023. Uncertainty-based dynamic economic dispatch for diverse load and wind profiles using a novel hybrid algorithm. *Environ. Dev. Sustain.* 25, 4723–4763. <https://doi.org/10.1007/S10668-022-02218-5/FIGURES/24>.
- Ben-Tal, A., El Ghaoui, L., Nemirovski, A., 2009. *Robust Optimization*. Princeton University Press.
- Ben-Tal, A., Goryashko, A., Guslitzer, E., Nemirovski, A., 2004. Adjustable robust solutions of uncertain linear programs. *Math. Program.* 99, 351–376. <https://doi.org/10.1007/S10107-003-0454-Y/METRICS>.
- Ben-Tal, A., Nemirovski, A., 1999. Robust solutions of uncertain linear programs. *Oper. Res. Lett.* 25, 1–13. [https://doi.org/10.1016/S0167-6377\(99\)00016-4](https://doi.org/10.1016/S0167-6377(99)00016-4).
- Bertsimas, D., Litvinov, E., Sun, X.A., Zhao, J., Zheng, T., 2013. Adaptive robust optimization for the security constrained unit commitment problem. *IEEE Trans. Power Syst.* 28, 52–63. <https://doi.org/10.1109/TPWRS.2012.2205021>.
- Birge, J.R., Louveaux, F., 2011. *Introduction to stochastic programming*. Springer Series in Operations Research and Financial Engineering. Springer, New York, New York, NY. <https://doi.org/10.1007/978-1-4614-0237-4>.
- Brás, M., Moura, A., Andrade-Campos, A., 2025. Cost efficiency in water supply systems: an applied review on optimization models for the pump scheduling problem. *Eur. J. Oper. Res.* 323, 1–19. <https://doi.org/10.1016/J.EJOR.2024.07.039>.
- Chacko, K., Alizadeh, M., Capitanescu, F., Li, P., 2024. Chance constrained framework for AC security constrained optimal power flow. In: *Proceedings of the 2024 International Conference on Smart Energy Systems and Technologies: Driving the Advances for Future Electrification, SEST 2024*. <https://doi.org/10.1109/SEST61601.2024.10694549>.
- Coffrin, C., Knueven, B., Holzer, J., Vuffray, M., 2021. The impacts of convex piecewise linear cost formulations on AC optimal power flow. *Electr. Power Syst. Res.* 199, 107191. <https://doi.org/10.1016/J.EPSR.2021.107191>.
- Coskun Dilcan, C., Aydinlal Koksal, M., 2025. A novel approach to resource management within the water-electricity-climate nexus perspective. *Sci. Total Environ.* 991, 179921. <https://doi.org/10.1016/J.SCIOTENV.2025.179921>.
- Creaco, E., Campisano, A., Fontana, N., Marini, G., Page, P.R., Walski, T., 2019. Real time control of water distribution networks: a state-of-the-art review. *Water Res.* 161, 517–530. <https://doi.org/10.1016/j.watres.2019.06.025>.
- Gao, Y., Sosnowski, M., Grabowska, K., Skrobek, D., Moeen Uddin, G., Kulakowska, A., Zylka, A., El Fil, B., Kuperman, A., Marzbani, F., Abdelfatah, A., 2024. Economic dispatch optimization strategies and problem formulation: a comprehensive review. *Energies* 17, 550. <https://doi.org/10.3390/EN17030550>, 2024, Vol. 17, Page 550.
- Gorissen, B.L., Yanikoglu, I., den Hertog, D., 2015. A practical guide to robust optimization. *Omega (Westport)* 53, 124–137. <https://doi.org/10.1016/J.OMEGA.2014.12.006>.
- Goryashko, A.P., Nemirovski, A.S., 2014. Robust energy cost optimization of water distribution system with uncertain demand. *Autom. Remote Control* 75, 1754–1769. <https://doi.org/10.1134/S000511791410004X>.
- Grosso, J.M., Velarde, P., Ocampo-Martinez, C., Maestre, J.M., Puig, V., 2017. Stochastic model predictive control approaches applied to drinking water networks. *Optim. Control Appl. Methods* 38, 541–558. <https://doi.org/10.1002/OCA.2269>.
- Housh, M., 2017. Non-probabilistic robust optimization approach for flood control system design. *Environ. Model. Softw.* 95, 48–60. <https://doi.org/10.1016/J.ENVSOF.2017.05.003>.
- Housh, M., 2011. *Optimal Multi-Year Management of Regional Water Resources Systems under Uncertainty*. Technion-Israel Institute of Technology.
- Housh, M., Ostfeld, A., Shamir, U., 2012. Optimal multi-year management of a regional water supply system under uncertainty: the affine adjustable robust counterpart (AARC) approach. In: *Proceedings of the World Environmental and Water Resources Congress 2012*, pp. 793–807. <https://doi.org/10.1061/9780784412312.082>.
- Crossing Boundaries, 2012 Congress.
- Housh, M., Ostfeld, A., Shamir, U., 2011. Optimal multiyear management of a water supply system under uncertainty: robust counterpart approach. *Water. Resour. Res.* 47, 1–15. <https://doi.org/10.1029/2011WR010596>.
- Irving, M.R., Sterling, M.J.H., 1987. Efficient Newton-Raphson algorithm for load-flow calculation in transmission and distribution networks. *IEE Proc. C (Gener. Transm. Distrib.)* 134, 325–328. <https://doi.org/10.1049/IP-C.1987.0053>.
- Jabr, R.A., 2013. Adjustable robust OPF with renewable energy sources. *IEEE Trans. Power Syst.* 28, 4742–4751. <https://doi.org/10.1109/TPWRS.2013.2275013>.
- Jamal, A., Linker, R., Housh, M., 2019. Optimal irrigation with perfect weekly forecasts versus imperfect seasonal forecasts. *J. Water Resour. Plan. Manage* 145, 06019003. [https://doi.org/10.1061/\(ASCE\)WR.1943-5452.0001066](https://doi.org/10.1061/(ASCE)WR.1943-5452.0001066).
- Jamal, A., Linker, R., Housh, M., 2018. Comparison of various stochastic approaches for irrigation scheduling using seasonal climate forecasts. *J. Water Resour. Plan. Manage.* 144, 04018028. [https://doi.org/10.1061/\(ASCE\)WR.1943-5452.0000951](https://doi.org/10.1061/(ASCE)WR.1943-5452.0000951).
- Jiang, H., Yao, L., Qin, J., Bai, Y., Brandt, M., Lian, X., Davis, S.J., Lu, N., Zhao, W., Liu, T., Zhou, C., 2025. Globally interconnected solar-wind system addresses future electricity demands. *Nat. Commun.* 16, 4523. <https://doi.org/10.1038/S41467-025-59879-9>.
- Jowitt, P.W., Germanopoulos, G., 1992. Optimal pump scheduling in water-supply networks. *J. Water Resour. Plan. Manage.* 118, 406–422. [https://doi.org/10.1061/\(ASCE\)0733-9496\(1992\)118:4\(406\)](https://doi.org/10.1061/(ASCE)0733-9496(1992)118:4(406)).
- Keyvandarian, A., Saif, A., 2024. An adaptive distributionally robust optimization approach for optimal sizing of hybrid renewable energy systems. *J. Optim. Theory. Appl.* 203, 2055–2082. <https://doi.org/10.1007/S10957-024-02518-Y/FIGURES/5>.
- Maier, H.R., Guillaume, J.H.A., van Delden, H., Riddell, G.A., Haasnoot, M., Kwakkel, J. H., 2016. An uncertain future, deep uncertainty, scenarios, robustness and adaptation: how do they fit together? *Environ. Model. Softw.* 81, 154–164. <https://doi.org/10.1016/J.ENVSOF.2016.03.014>.
- Mieth, R., Dvorkin, Y., 2018. Data-driven distributionally robust optimal power flow for distribution systems. *IEEE Control Syst. Lett.* 2, 363–368. <https://doi.org/10.1109/LCSYS.2018.2836870>.
- Molinos-Senante, M., Maziotis, A., Sala-Garrido, R., Mocholi-Arce, M., 2022. Understanding water-energy nexus in water provision: an eco-efficiency assessment of water companies. *Water Res.* 225, 119133. <https://doi.org/10.1016/J.WATRES.2022.119133>.
- Molzahn, D.K., Roald, L.A., 2018. Towards an AC optimal power flow algorithm with robust feasibility guarantees. In: *Proceedings of the 20th Power Systems Computation Conference, PSCC 2018*. <https://doi.org/10.23919/PSCC.2018.8442889>.
- Naval, N., Yusta, J.M., 2021. Optimal short-term water-energy dispatch for pumping stations with grid-connected photovoltaic self-generation. *J. Clean. Prod.* 316, 128386. <https://doi.org/10.1016/J.JCLEPRO.2021.128386>.
- Nourianfar, H., Abdi, H., 2023. Economic emission dispatch considering electric vehicles and wind power using enhanced multi-objective exchange market algorithm. *J. Clean. Prod.* 415, 137805. <https://doi.org/10.1016/J.JCLEPRO.2023.137805>.
- Oikonomou, K., Parvania, M., 2020. Optimal coordinated operation of interdependent power and water distribution systems. *IEEE Trans. Smart Grid* 11, 4784–4794. <https://doi.org/10.1109/TSG.2020.3000173>.
- Ormsbee, L.E., Lansey, K.E., 1994. Optimal control of water supply pumping systems. *J. Water Resour. Plan. Manage.* 120, 237–252. [https://doi.org/10.1061/\(ASCE\)0733-9496\(1994\)120:2\(237\)](https://doi.org/10.1061/(ASCE)0733-9496(1994)120:2(237)).
- Pankaj, B.S., Jaykrishnan, G., Ostfeld, A., 2022. Optimizing Water quality treatment levels for water distribution systems under mixing uncertainty at junctions. *J. Water Resour. Plan. Manage.* 148, 04022013. [https://doi.org/10.1061/\(ASCE\)WR.1943-5452.0001544](https://doi.org/10.1061/(ASCE)WR.1943-5452.0001544).
- Perelman, G., Ostfeld, A., 2023. Adjustable robust optimization for water distribution system operation under uncertainty. *Water Resour. Res.* 59, e2023WR035508. <https://doi.org/10.1029/2023WR035508>.
- Perelman, G., Ostfeld, A., Fishbain, B., 2023. Robust optimal operation of water distribution systems. *Water* 15, 963. <https://doi.org/10.3390/W15050963>, 2023, Vol. 15, 963.
- Perelman, L., Housh, M., Ostfeld, A., 2013. Robust optimization for water distribution systems least cost design. *Water Resour. Res.* 49, 6795–6809. <https://doi.org/10.1002/wrcr.20539>.
- Portilla-Paveri, M., Cariaga, D., Negrete-Pincetic, M., Lorca, Á., Anjos, M.F., 2024. A long-term generation and transmission expansion planning model considering desalination flexibility and coordination: a Chilean case study. *Appl. Energy* 371, 123578. <https://doi.org/10.1016/J.APENERGY.2024.123578>.
- Salomons, E., Cao, H., Korder, K., Ostfeld, A., Li, P., 2025. Real-Time optimal operation of water systems under demand uncertainty and maximum water age constraints. *Water Resour. Res.* 61, e2024WR038587. <https://doi.org/10.1029/2024WR038587>.
- Salomons, E., Housh, M., 2020a. A practical optimization scheme for real-time operation of water distribution systems. *J. Water Resour. Plan. Manage.* 146, 1–12. [https://doi.org/10.1061/\(ASCE\)WR.1943-5452.0001188](https://doi.org/10.1061/(ASCE)WR.1943-5452.0001188).
- Salomons, E., Housh, M., 2020b. Practical real-time optimization for energy efficient water distribution systems operation. *J. Clean. Prod.* 275, 124148. <https://doi.org/10.1016/j.jclepro.2020.124148>.
- Salomons, E., Housh, M., Katz, D., Sela, L., 2023. Water-energy nexus in a desalination-based water sector: the impact of electricity load shedding programs. *npj Clean Water* 6 (1), 1–9. <https://doi.org/10.1038/s41545-023-00281-7>, 2023.
- Schwartz, Rafael, Housh, M., Ostfeld, A., 2016. Limited multistage stochastic programming for water distribution systems optimal operation. *J. Water Resour. Plan. Manage.* 142, 06016003. [https://doi.org/10.1061/\(ASCE\)WR.1943-5452.0000687](https://doi.org/10.1061/(ASCE)WR.1943-5452.0000687).
- Schwartz, Rafael, Housh, M., Ostfeld, A., 2016. Least-cost robust design optimization of water distribution systems under multiple loading. *ascelibrary.org* 142. [https://doi.org/10.1061/\(ASCE\)WR.1943-5452.0000670](https://doi.org/10.1061/(ASCE)WR.1943-5452.0000670).
- Sharma, S., Li, Q., 2024. Decentralized optimization of energy-water nexus based on a mixed-integral boundary compatible algorithm. *Appl. Energy* 359, 122588. <https://doi.org/10.1016/J.APENERGY.2023.122588>.
- Shmaya, T., Housh, M., Pecci, F., Baker, K., Kasprzyk, J., Ostfeld, A., 2024. Conjunctive optimal operation of water and power networks. *Heliyon* 10, e39136. <https://doi.org/10.1016/J.HELIYON.2024.E39136>.
- Sowby, R.B., 2023. Making waves: research to support water and wastewater utilities in the transition to a clean-energy future. *Water Res.* 233, 119739. <https://doi.org/10.1016/J.WATRES.2023.119739>.

- Stuhlmacher, A., Mathieu, J.L., 2022. Uncertainty-aware methods for leveraging water pumping flexibility for power networks.
- Stuhlmacher, A., Mathieu, J.L., 2020a. Water distribution networks as flexible loads: a chance-constrained programming approach. *Electr. Power Syst. Res.* 188, 106570. <https://doi.org/10.1016/J.EPSR.2020.106570>.
- Stuhlmacher, A., Mathieu, J.L., 2020b. Chance-constrained water pumping to manage water and power demand uncertainty in distribution networks. *Proc. IEEE* 108, 1640–1655. <https://doi.org/10.1109/JPROC.2020.2997520>.
- Szinai, J.K., Yates, D., Sánchez-Pérez, P.A., Staadecker, M., Kammen, D.M., Jones, A.D., Hidalgo-Gonzalez, P., 2024. Climate change and its influence on water systems increases the cost of electricity system decarbonization. *Nat. Commun.* 15, 1–14. <https://doi.org/10.1038/S41467-024-54162-9>;TECHMETA.
- Taheri, B., Molzahn, D.K., 2024. Optimizing parameters of the DC power flow. *Electr. Power Syst. Res.* 235, 110719. <https://doi.org/10.1016/J.EPSR.2024.110719>.
- Tan, T., Yang, M., Xie, R., Cao, Y., Chen, Y., 2025. Adjustable robust optimization with decision-dependent uncertainty for power system problems: a review. *Energy Convers. Econ.* 6, 155–169. <https://doi.org/10.1049/ENC2.70010>.
- Thomas, M., Sela, L., 2024. A mixed-integer linear programming framework for optimization of water network operations problems. *Water Resour. Res.* 60, e2023WR034526. <https://doi.org/10.1029/2023WR034526>.
- Tinney, W., Hart, C.E., 1967. Power flow solution by newton's method. *IEEE Trans. Power Appar. Syst.* 1449–1460. <https://doi.org/10.1109/TPAS.1967.291823>. PAS-86.
- Toba, A.L., Boire, L.D., Roni, M., 2024. Interdependent water and power infrastructure model (IWPIM): a modeling approach for water and energy resource management in rural communities. *Heliyon* 10, e32122. <https://doi.org/10.1016/J.HELIYON.2024.E32122>.
- Wang, Y., Xiong, G., 2025. Metaheuristic optimization algorithms for multi-area economic dispatch of power systems: part I—a comprehensive survey. *Artif. Intell. Rev.* 58, 1–48. <https://doi.org/10.1007/S10462-024-11070-0>;FIGURES/17.
- Wen, Y., Shen, L., Zhou, H., Wei, M., 2024. A real-time dynamic economic dispatch method for integrated power generation system considering the impact of high proportion of new energy generation. In: *Proceedings of the 2024 3rd International Conference on Energy, Power and Electrical Technology, ICEPET*, 2024, pp. 155–161. <https://doi.org/10.1109/ICEPET61938.2024.10626196>.
- Wu, S., Wang, J., Xu, H., Zhong, Y., Rao, J., 2025. End-to-end stochastic predict-then-optimize for cost-efficient water-energy resource scheduling. *IEEE Trans. Smart Grid.* <https://doi.org/10.1109/TSG.2025.3601238>.
- Wu, Z., Zhao, X., Mao, Y., Liu, R., 2024. Flexibility quantification of desalination plants for demand response: an adaptive robust optimization methodology. *Appl. Energy* 373, 123835. <https://doi.org/10.1016/J.APENERGY.2024.123835>.
- Xie, R., Pinson, P., Xu, Y., Chen, Y., 2024. Robust generation dispatch with purchase of renewable power and load predictions. *IEEE Trans. Sustain. Energy* 15, 1486–1501. <https://doi.org/10.1109/TSTE.2024.3353779>.
- Yan, Q., Zhang, M., Lin, H., Li, W., 2022. Two-stage adjustable robust optimal dispatching model for multi-energy virtual power plant considering multiple uncertainties and carbon trading. *J. Clean. Prod.* 336, 130400. <https://doi.org/10.1016/J.JCLEPRO.2022.130400>.
- Yin, X., Zhao, Z., Yang, W., 2023. Optimizing cleaner productions of sustainable energies: a co-design framework for complementary operations of offshore wind and pumped hydro-storages. *J. Clean. Prod.* 396, 135832. <https://doi.org/10.1016/J.JCLEPRO.2022.135832>.
- Zhao, X., Liu, D., Wu, Z., Huang, C., 2024. Chance constrained optimal power-water flow: an iterative algorithm assisted by deep neural networks. *Electr. Power Syst. Res.* 234, 110599. <https://doi.org/10.1016/J.EPSR.2024.110599>.
- Zhou, Y., Hou, H., Yan, H., Wang, X., Zhou, R., 2025. Data-driven distributionally robust stochastic optimal dispatching method of integrated energy system considering multiple uncertainties. *Energy* 325, 136104. <https://doi.org/10.1016/J.ENERGY.2025.136104>.